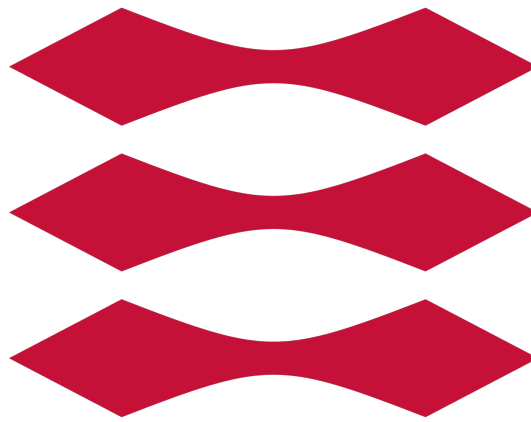


Danish Technical University
**41685 Materials Characterization and
Testing summary**

Comprehensive Study Notes from Spring 2026

DTU



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1 Sample preparation

Purpose: To make the material suitable for observation in Light Optical Microscopy, Secondary Electron Microscopy, hardness testing, or other characterization methods.

The key goal is,

Reveal the real structure and avoid artefacts

Where an artefact is something that looks like a real microstructure feature but was actually created during preparation. Examples include scratches, smeared material, pull-out, deformation, cracks from cutting, edge rounding, or polishing relief.

The usual preparation sequence is as follows,

Sectioning → Mounting → Grinding → Polishing → Etching

1.1 Sectioning

Sectioning refers to cutting a piece from the material so that the area of interest can be examined.

Purpose: To expose the correct cross or longitudinal-section for surface for analysis.

For example you may section,

- Across a coating to measure coating thickness
- Through a weld to examine heat-affected zones
- Through a fracture surface region
- Across a surface treated layer
- Through a cast component to look for porosity of inclusions

Sectioning must be done carefully because cutting can introduce damage. The most common problems are,

heat + mechanical deformation + cracking

If the sample gets too hot during cutting, the microstructure can change. This is especially important for heat-treated metals, polymers or materials with phase transformations.

For metals, overheating can cause tempering, oxidation, recrystallization, or local softening/hardening. For brittle materials, aggressive cutting can create cracks.

How to avoid artefacts

- Low cutting force
- Sufficient cooling
- Suitable cutting wheel/blade
- Slow enough feed rate
- Minimal vibration

1.2 Mounting

Mounting refers to embedding the sample in a material, usually resin, so it becomes easier to handle during grinding and polishing.

Purpose: Mounting helps with,

- Easier handling
- Protecting edges
- Keeping small or irregular samples stable
- Allowing automated grinding/polishing
- Producing a flat surface
- Supporting fragile samples

Thus is especially important if the sample is small, thin, brittle, porous, coated or oddly shaped.

Types of mounting

There are generally two main types of mounting:

Hot mounting, where the sample is mounted under heat and pressure, often using thermosetting or thermoplastic resin.

Advantages:

- Fast
- Strong support
- Good for routine metallic

Limitations:

- Not suitable for heat sensitive materials
- Pressure can damage fragile or porous samples

Cold mounting, where the sample is embedded in resin at room temperature, usually epoxy, acrylic, or polyester.

Advantages:

- Better for fragile, porous, or heat sensitive samples
- Useful for coatings and delicate samples
- Epoxy can penetrate pores or cracks better

Limitations:

- Slower curing
- May shrink depending on resin
- Poor impregnation can leave gaps

Important artefact: edge rounding

One of the biggest mounted-related issues is edge rounding. If the sample edge is not well supported, the edge can be polished away more than the center. This is a problem when studying coatings, surface treatments, carburized/nitride layers, corrosion layers, or thin films

1.3 Grinding

Grinding removes the material using abrasive papers or discs. It removes cutting damage and makes the sample flat.

Purpose:

Remove sectioning damage + Create a flat surface

The surface after cutting is usually rough and damaged. grinding removes that damaged layer.

How grinding works

Grinding uses abrasive particles that cut grooves into the surface. coarse grinding removed material quickly but leaves deep scratches. Fine grinding removed the deep scratches from the previous step.

This is why grinding is done in steps,

coarse $\rightarrow$ medium $\rightarrow$ fine

Key rule, each grinding step must remove the scratches from the previous step.

A common practical method is to rotate the sample by 90° between steps. This makes it easier to see whether from the previous step have been removed.

Common grinding artefacts

Scratches these are grooves from abrasive particles. If not removed they can be mistaken for cracks or deformation lines

Embedded abrasive particles Abrasive materials can become stuck in soft materials

Smearing soft phases may be dragged across the surface, covering harder phases or pores.

Pull-out Hard particles, inclusions, or brittle phases may be pulled out, leaving holes that can be mistaken for porosity.

Deformation Grinding can plastically deform the surface, especially in soft metals.

Edge rounding Excessive grinding or poor mounting can round sample edges.

1.4 Polishing

Polishing removes the fine scratches left from grinding and produces a smooth surface suitable for microscopy.

Purpose:

Remove fine scratches + Remove deformation layer + Prepare for etching

A polished surface should be flat and free from scratches, deformation and contamination. The purpose is not only to make the sample look shiny, but to make sure that the surface represents the real microstructure.

Mechanical polishing

Mechanical polishing uses fine abrasive particles, often diamond suspension, alumina, or silica based suspensions.

Typical polishing can be done in progressively finer steps, for example,

$$3\,\mu\text{m} \rightarrow 1\,\mu\text{m} \rightarrow 40\,\text{nm OPS}$$

Where OPS is an oxide polishing suspension, often colloidal silica. OPS is useful as a final polishing step because it can remove the last damaged layer and improve the surface before microscopy or etching.

Common polishing artefacts

Relief can occur when different phases polish at different rates. Softer phases may be removed faster than harder phases, producing height differences.

Smearing can occur when soft material is dragged across the surface and covers harder phases, pores, or inclusions.

Pull-out can occur when hard particles, brittle phases, carbides, graphite, or inclusions are removed from the matrix, leaving holes.

Scratches can still appear during polishing if the cloth is contaminated by larger abrasive particles.

Over-polishing can remove real surface features, round edges, or exaggerate relief.

Thus, polishing should remove the damage from grinding, but it should not introduce new artefacts that could be mistaken for the real microstructure.

1.5 Etching

Etching is used to reveal the microstructure after polishing. A polished metal surface can often look almost featureless because it reflects light uniformly.

Purpose: Etching creates contrast by attacking different parts of the microstructure differently.

Etching can reveal,

- Grain boundaries
- Different phases
- Segregation
- Deformation structures
- Carbides
- Weld zones
- Heat-treated structures
- Surface layers

How etching works

Etching works because different microstructural regions have different chemical reactivity or crystallographic orientation.

For example,

$$\begin{aligned}\text{grain boundary} &\neq \text{grain interior} \\ \text{phase A} &\neq \text{phase B}\end{aligned}$$

Therefore, the etchant attacks some regions more than others, creating contrast in reflected light microscopy.

Types of etching

Chemical etching is where the sample is exposed to a chemical reagent for a controlled time. The etchant must be chosen based on the material and on the feature that should be revealed.

Electrolytic etching is where an electrical potential is applied while the sample is in an electrolyte. This can be useful for specific alloys or phases.

Stain etching creates colour contrast and can sometimes be enhanced using polarized light.

Under-etching and over-etching

Under-etching means that the sample has not been etched enough. The microstructure will have poor contrast and grain boundaries or phases may be difficult to see.

Over-etching means that the sample has been etched too much. The surface can become too rough, dark, or attacked, and fine details may be lost.

Etching should therefore be controlled, because too little etching gives poor contrast, while too much etching can destroy or exaggerate the microstructure.

1.6 Real structure vs artefacts

The main challenge in sample preparation is to reveal the real structure and prevent artefacts.

Real microstructural features

Examples of real microstructural features are,

- Grain boundaries
- Pores
- Cracks
- Inclusions
- Carbides
- Different phases
- Coating layers
- Deformation bands
- Precipitates
- Heat-affected zones

Preparation artefacts

Examples of preparation artefacts are,

- Scratches from grinding or polishing
- Smearing of soft phases
- Pull-out of particles or brittle phases
- Edge rounding
- Polishing relief
- Deformation layer from cutting or grinding
- Cracks from cutting
- Oxidation from overheating
- Contamination from polishing media
- Over-etching or under-etching

Many artefacts can look like real microstructural features. For example, scratches may look like cracks, pull-out may look like porosity, and polishing relief may exaggerate phase contrast. Therefore, microscopy images should always be interpreted critically.

1.7 Sample preparation and light microscopy contrast

Sample preparation affects which light microscopy contrast mechanisms can be used.

The most common reflected light microscopy methods are,

- Bright-field illumination
- Dark-field illumination
- Oblique illumination
- Differential interference contrast, DIC
- Polarized light

Bright field

Bright field is the most common method. Light is reflected approximately perpendicular to the sample surface. Flat matrix regions usually appear bright, while grain boundaries, pores, cracks, and etched features often appear darker.

Good preparation is important because scratches and polishing damage can also become visible and may be confused with real features.

Dark field

In dark field, features that scatter light appear bright. This means that raised particles, edges, cracks, pores, carbides, or surface relief can become more visible.

Dark field is therefore useful for observing features that scatter light, but it is also sensitive to surface relief and polishing artefacts.

Oblique illumination

Oblique illumination uses angled light to create shadows. This can help show whether features are raised or recessed.

This is useful for interpreting polishing relief, pits, cracks, or topographical features.

Differential interference contrast, DIC

DIC enhances small height differences on the sample surface. It can make etched microstructures and small surface relief easier to see.

Polarized light

Polarized light can enhance contrast in anisotropic materials or when stain etchants are used. It is useful when normal bright field does not give enough contrast.

Thus, the quality of sample preparation determines which contrast mechanisms can be used reliably. A flat polished surface is needed for reliable bright-field imaging, while relief or etched topography can be emphasized using dark field, oblique illumination, or DIC.

1.8 Why preparation matters for different materials

Different materials respond differently to sectioning, grinding, polishing and etching.

Metals

Metals usually need grinding, polishing and etching. Etching is often necessary to reveal grains, phases, and heat treatment effects.

Soft metals can smear during grinding or polishing, while hard particles may lead to pull-out or polishing relief.

Ceramics

Ceramics are brittle and can crack or chip during sectioning and grinding. They may also suffer from grain pull-out. Therefore, careful cutting and polishing are important.

Polymers

Polymers are often soft and heat sensitive. Cutting and grinding can cause smearing, melting, or deformation if the preparation is too aggressive.

Coatings

Coatings require good mounting and edge support. Otherwise, the coating can crack, delaminate, or be rounded at the edge.

This is especially important when measuring coating thickness or studying surface treatments.

Porous materials

Porous materials may need impregnation with resin before grinding and polishing. Otherwise pores can collapse, fill with polishing debris, or cause pull-out.

1.9 Sample preparation for hardness testing

For Vickers hardness testing, the surface must be,

flat + smooth + polished enough to see the diagonals

Bad preparation gives unreliable hardness values.

Examples of preparation problems are,

- A rough surface makes the indentation diagonals difficult to measure
- A tilted surface gives a distorted indentation
- A deformed surface layer from grinding can affect the measured hardness
- Coating/substrate effects can affect measurements on thin layers
- Indents near edges, pores, cracks, or other indents are unreliable

Sample preparation is therefore important for Vickers hardness because the indentation diagonals must be measured accurately.

1.10 Sample preparation for SEM and EDS

The same preparation logic also applies to SEM and SEM-EDS.

For SEM, the surface must represent the real structure. Scratches, contamination, pull-out, and polishing relief can all affect image interpretation.

Important SEM preparation aspects are,

- The surface should be clean and representative
- Loose particles should be removed
- Non-conductive samples may need conductive coating
- Charging should be avoided because it can distort the image
- The preparation should not create artificial cracks, pores, or deformation

For EDS, preparation is especially important because EDS detects elements. Contamination or preparation residues can therefore appear in the spectrum.

Important EDS preparation aspects are,

- Polishing residues can give misleading elemental peaks
- Oxidation can increase the measured oxygen signal
- Resin can contribute carbon or oxygen
- Carbon coating can affect carbon measurements

- Thin coatings can be difficult to analyse because the interaction volume may include the substrate
- Small phases can be difficult to analyse because the signal may include the surrounding material

Therefore, for SEM-EDS, sample preparation must avoid contamination and must be chosen based on the size of the features that should be analysed.

2 Light Optical Microscopy

Light Optical Microscopy, LOM, uses visible light and lenses to magnify and image the sample. In materials characterization, LOM is commonly used in **reflected light mode**, because many engineering materials such as metals, ceramics and coatings are opaque. In reflected light microscopy, the light is sent through the objective, reflected from the sample surface, and collected back through the same objective.

LOM is often the first characterization method used because it gives a quick overview of the microstructure. It can be used to identify features such as grains, phases, inclusions, pores, cracks, coatings, corrosion, deformation structures and surface defects. However, it is limited by the wavelength of visible light and therefore has lower resolution than electron microscopy.

The main idea is,

LOM gives an overview of the microstructure, but depends strongly on sample preparation and optical contrast

Light microscopy should therefore be used to put forward hypotheses and define which further characterization techniques may be needed, such as SEM, EDS or XRD.

2.1 Principle of light optical microscopy

In LOM, the image is formed using light that interacts with the sample. The contrast in the image comes from differences in how the material reflects, absorbs, refracts, diffracts, scatters or changes the polarization of light.

For reflected light microscopy, contrast can come from,

reflection + absorption + surface topography + phase differences + polarization effects

Different phases or grains may reflect light differently. Etched grain boundaries may scatter light and appear dark. Surface relief can create shadows. Thin films or oxides can create interference colours. Therefore, the observed image is not just a direct picture of the material, but a result of light-matter interaction.

2.2 Light as a wave

Light can be described as an electromagnetic wave. The important quantities are wavelength, frequency, velocity and energy.

The photon energy is given by,

$$E = \frac{hc}{\lambda}$$

where E is the photon energy, h is Planck's constant, c is the speed of light and λ is the wavelength.

The angular frequency is,

$$\omega = 2\pi f$$

The wave number is,

$$k = \frac{2\pi}{\lambda}$$

These relations explain why wavelength matters in microscopy. The wavelength determines the diffraction behaviour of light and therefore limits the resolution of the microscope.

The key point is,

$$\lambda \downarrow \Rightarrow \text{better possible resolution}$$

2.3 Refractive index

The refractive index describes how much the speed of light is reduced inside a material compared with vacuum.

$$n = \frac{c_0}{v}$$

where n is the refractive index, c_0 is the speed of light in vacuum and v is the speed of light in the material.

Since light is an electromagnetic wave, the refractive index depends on how the material responds to the electric and magnetic fields of the wave. The refractive index also depends on wavelength, which is called **dispersion**.

The refractive index is important because it affects,

$$\text{refraction} + \text{reflection} + \text{numerical aperture} + \text{resolution}$$

A higher refractive index immersion medium can increase the numerical aperture and therefore improve resolution.

2.4 Refraction and Snell's law

When light passes from one medium to another with a different refractive index, its direction changes. This is called refraction. The relation between the angles and refractive indices is given by Snell's law,

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

where n_1 and n_2 are the refractive indices, and θ_1 and θ_2 are the angles relative to the normal.

For microscopy, refraction matters because lenses work by bending light. Refraction at curved lens surfaces makes light converge or diverge, allowing the microscope to form an image.

2.5 Reflection and absorption

In reflected light microscopy, the light that reaches the detector is light reflected from the sample surface. Different materials reflect different amounts of light.

Metals generally have high reflectivity because they contain many free electrons. Different metals can also have different colours because they absorb different wavelengths of light.

This is why different phases, coatings or oxides may appear with different brightness or colour in LOM.

$$\text{high reflectivity} \Rightarrow \text{bright region}$$

$$\text{low reflectivity or strong absorption} \Rightarrow \text{dark region}$$

However, brightness should not be interpreted blindly. A dark feature may be a grain boundary, pore, crack, oxide, etched phase, shadow or preparation artefact.

2.6 Interference and colours

Interference occurs when two light waves overlap. If the waves are in phase, they interfere constructively. If they are out of phase, they interfere destructively.

A phase shift can be introduced by a difference in optical path length. This means that constructive and destructive interference can create bright and dark regions.

Interference is important in microscopy because thin films can create colours. This happens when reflected waves from different interfaces interfere constructively or destructively depending on film thickness and wavelength.

Examples where interference colours can occur are,

- Liquid films
- Solid thin films
- Oxide layers
- Crystals
- Air gaps

Therefore, thin-film colours are not always due to chemical colour. They can also come from optical interference.

2.7 Diffraction

Diffraction is the spreading of light when it interacts with an object or aperture. In microscopy, diffraction is very important because it limits the resolution.

The important idea is,

small features in the sample plane $\Rightarrow$ large diffraction angles

If the objective does not collect enough diffracted light, the fine detail cannot be reconstructed in the image. This is why numerical aperture is important.

Diffraction is both necessary and limiting. It is necessary because sample features diffract light and create image information, but it also limits resolution because the objective can only collect light within a certain angle.

2.8 Image formation in a microscope

A microscope forms an image by collecting light from the sample and focusing it through the objective and eyepiece or camera system.

In a compound microscope, the objective first forms a real intermediate image. The eyepiece then allows the eye to view a magnified virtual image. In a camera-based microscope, the recorded image is based on the intermediate image formed by the microscope optics.

For a positive thin lens, the lens equation is,

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}$$

where s is the object distance, s' is the image distance and f is the focal length.

The lateral magnification is,

$$M = \frac{h_i}{h_o} = -\frac{s'}{s}$$

where h_i is the image height and h_o is the object height.

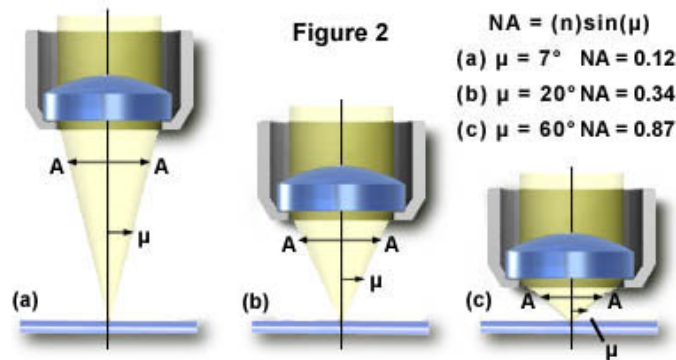
The negative sign means that the image is inverted.

2.9 Numerical aperture

The numerical aperture, NA, describes the range of angles over which the objective can collect light from the sample.

$$NA = n \sin \alpha$$

where n is the refractive index of the medium between sample and objective, and α is the collection half-angle.



A higher NA means that the objective collects light over a larger angle. This gives,

higher resolution + more collected light + smaller depth of field

Numerical aperture is one of the most important parameters in microscopy. A higher NA improves resolution because more diffracted light from fine features is captured.

2.10 Immersion medium

An immersion medium, such as oil, can be used between the objective and sample. This increases the refractive index n in front of the objective.

Since,

$$NA = n \sin \alpha$$

increasing n increases the numerical aperture. Oil immersion therefore improves resolution because it allows the objective to collect more diffracted light from the sample.

2.11 Resolution

Resolution is the ability to distinguish two close points as separate objects. It is not the same as magnification.

The Rayleigh criterion gives the minimum resolvable distance as,

$$d = \frac{0.61\lambda}{NA}$$

where d is the minimum resolved distance, λ is the wavelength and NA is the numerical aperture.

A smaller d means better resolution. Therefore, resolution improves when,

$$\lambda \downarrow$$

or,

$$NA \uparrow$$

This is a very important point,

$$\text{Magnification} \neq \text{Resolution}$$

Magnification only makes the image larger. Resolution determines whether two close features can actually be separated. Increasing magnification without improving resolution only gives a larger blurry image.

2.12 Abbe diffraction limit

Abbe's theory explains resolution using diffraction from the sample. A sample can be considered as a superposition of diffraction gratings. Fine features diffract light at large angles. To form an image of these fine features, the objective must collect enough diffraction orders.

A common expression of the Abbe diffraction limit is,

$$d = \frac{\lambda}{2NA}$$

This means that the microscope cannot resolve features if the diffracted light from those features is not collected by the objective. Small features diffract at high angles, so a high NA objective is needed to capture the diffraction information.

2.13 Point spread function

The point spread function, PSF, describes how a single point object is imaged by the microscope. Because of diffraction, a point is not imaged as a perfect point, but as a small diffraction pattern called an Airy pattern.

The optical image can therefore be seen as a superposition of many PSFs. If two points are too close, their PSFs overlap too much and they cannot be distinguished.

The main idea is,

$$\text{smaller PSF} \Rightarrow \text{better resolution}$$

2.14 Rayleigh criterion

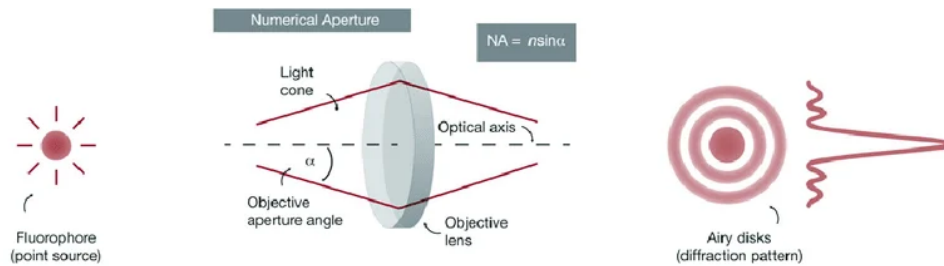
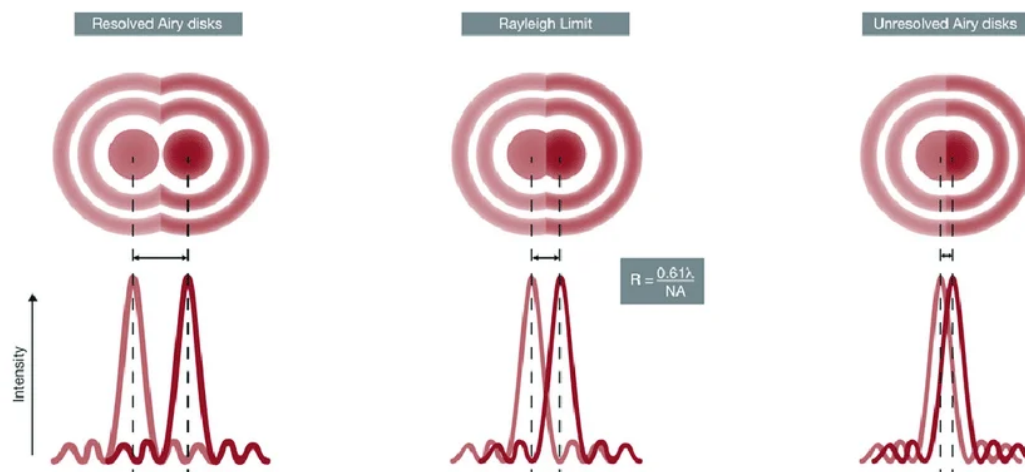
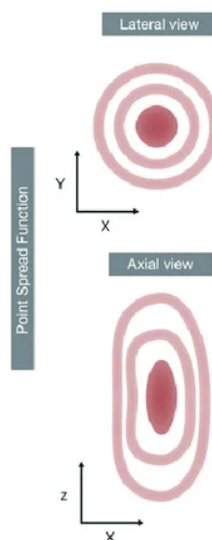
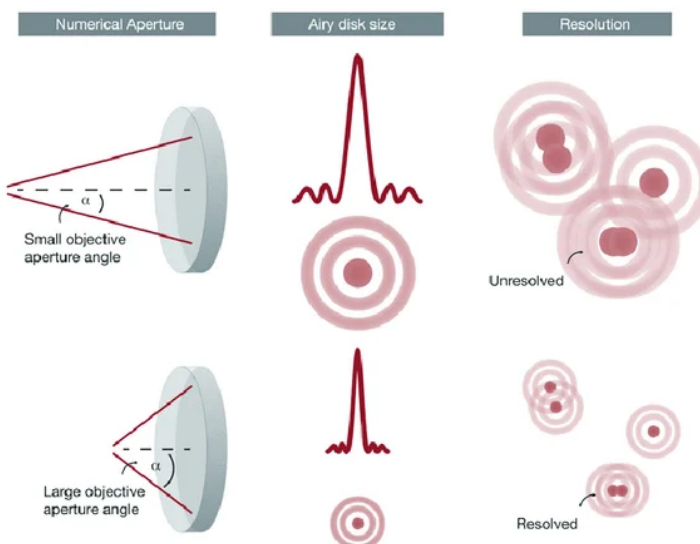
The Rayleigh criterion defines when two point objects are just resolved. Two points are considered resolved when the central maximum of one Airy disk overlaps with the first minimum of the other.

The key relation is,

$$r \geq d = \frac{0.61\lambda}{NA}$$

where r is the distance between two points.

This criterion gives a practical definition of resolution. Two points are just resolved when their diffraction patterns overlap in a way that still leaves a visible intensity dip between them.

A Numerical Aperture (NA)**B** Resolution (R)**C** Point Spread Function (PSF)**D** Numerical Aperture, Airy disk size and resolution**2.15 Contrast**

Contrast is the difference in intensity or colour between different regions of the image. Without contrast, even a high-resolution microscope cannot show useful detail.

For materials, contrast can come from,

- Grain boundaries

- Different phases
- Pores and cracks
- Surface relief
- Thin films or oxides
- Etching response
- Polarization effects

Good resolution is not useful if there is no contrast. Therefore, sample preparation, etching and illumination mode are important for LOM interpretation.

2.16 Bright-field illumination

Bright-field illumination is the most common reflected light microscopy method. The light reflection is mainly perpendicular to the specimen surface. In bright field, flat matrix regions usually appear bright, while grain boundaries, pores, cracks or etched features appear dark.

This is because flat regions reflect light back into the objective, while inclined, rough or etched regions scatter light away.

2.17 Dark-field illumination

In dark-field illumination, direct reflected light is not collected in the same way as in bright field. Instead, light scattered by surface features enters the objective.

This means that features that are dark in bright field can appear bright in dark field. Dark field is useful for features that scatter light, such as edges, particles, carbides, cracks or surface relief.

2.17.1 Difference in Bright-field and Dark-field

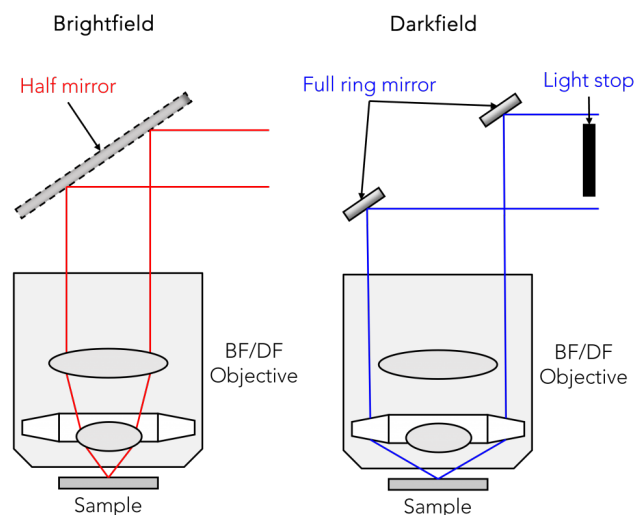


Figure 1: Differences between reflected bright-field and reflected dark-field within the microscope.

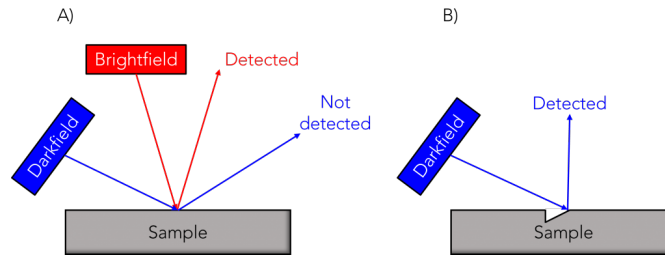


Figure 2: A) Differences in the incident angle of brightfield and darkfield and B) the detection of defect scattered light in darkfield.

2.18 Oblique illumination

Oblique illumination uses light that is de-centred or angled relative to the optical axis. This produces shadows on microstructural features.

Oblique illumination enhances topography by creating shadows. It can help identify whether features are raised or recessed, but the illumination direction must be known to interpret the image correctly.

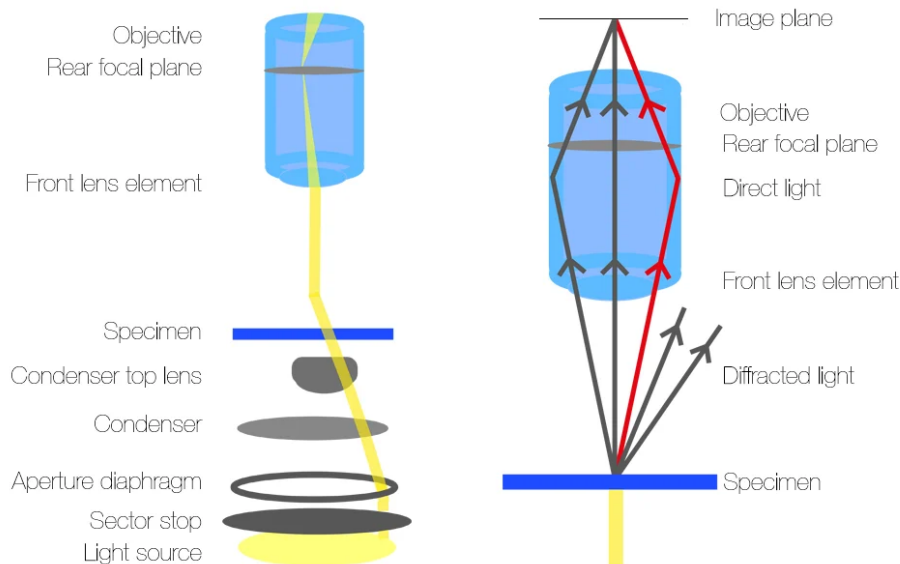


Figure 3: 1a (left) shows a typical oblique contrast configuration which causes the light to hit the specimen at an angle. Figure 1b (right) demonstrates that only the diffracted sidebands of light on one half of the direct beam are collected by the objective, which introduces contrast into the image.

2.19 Differential interference contrast

Differential interference contrast, DIC, uses optical components such as a beam-splitting prism, polarizer and analyzer. It produces contrast from small height differences and phase differences in the reflected light.

DIC is useful for revealing small surface height differences or polishing relief. It gives a pseudo-3D appearance, but it should not be interpreted as an actual height measurement unless calibrated.

2.20 Polarized light

Polarized light microscopy uses polarizers to control the polarization direction of light. Some materials change the polarization of reflected light depending on crystallographic orientation, anisotropy or surface condition.

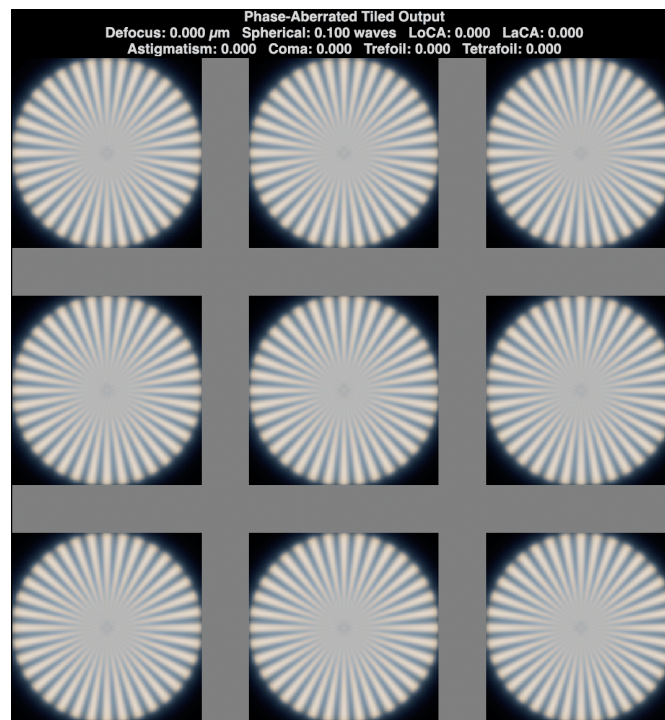
Polarized light is useful when the sample changes the polarization of reflected light. It can enhance contrast between grains, phases or stain-etched regions, especially in anisotropic materials.

2.21 Lens aberrations

Real lenses are not ideal. They do not focus all rays perfectly to the same point. This can reduce sharpness, distort the image or introduce colour errors.

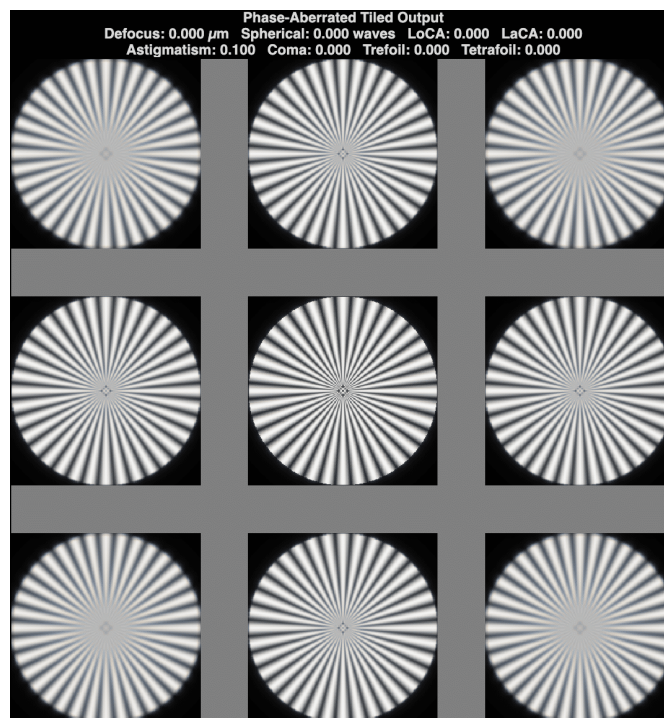
Spherical aberration

Spherical aberration occurs when rays passing through different parts of the lens focus at different points. This makes the image less sharp.



Astigmatism

Astigmatism occurs when the lens focuses rays differently in different directions. This can make points appear stretched or blurred.

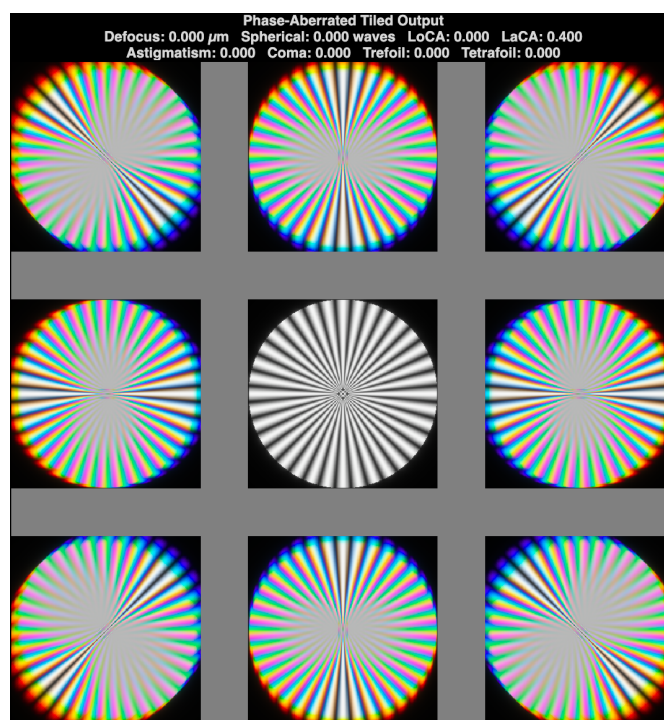


Distortion

Distortion changes the geometry of the image. Straight lines may appear curved or distances may not be represented correctly.

Chromatic aberration

Chromatic aberration occurs because different wavelengths are refracted by different amounts. This means different colours focus at different positions.



Achromatic lenses correct mainly for red and blue wavelengths, while apochromatic lenses correct better over red, green and blue wavelengths.

2.22 Depth of field

Depth of field is the vertical range over which the sample remains acceptably in focus. In microscopy, high magnification and high numerical aperture usually give a smaller depth of field.

The important practical point is,

$$NA \uparrow \Rightarrow \text{resolution improves, but depth of field decreases}$$

This is why rough surfaces can be difficult to image fully in focus using conventional light microscopy.

2.23 Digital microscopy

Digital microscopy uses a camera and software to record and process images. It can be used for image capture, measurement and extended-depth-of-field imaging.

Digital microscopy can be useful for measuring,

- Lengths
- Areas
- Angles
- Pit sizes
- Particle sizes
- Layer thicknesses
- Surface features

Extended-depth-of-field images can be made by taking several images at different focus heights and combining them into one image. This is useful for rough surfaces or 3D features.

However, the measurements still depend on calibration, resolution and correct focusing.

2.24 Digital resolution and pixel size

Even if the optical microscope can resolve a feature, the camera must sample it properly. The camera pixel size must be adjusted to the optical resolution.

This is related to the Nyquist limit, where at least about two pixels are needed across the smallest resolvable feature.

A useful relation is,

$$M \cdot \text{pixel size} = 2r = 2 \cdot \frac{0.6\lambda}{NA}$$

The exact form depends on how the pixel size is defined, but the main idea is that the detector must sample the optical image finely enough.

If the pixel size is too large, the detector can lose information even if the microscope optics could resolve it.

2.25 What LOM is good for

LOM is useful for,

- Overview of microstructure

- Grain size and grain shape
- Phase distribution
- Porosity and cracks
- Coating thickness
- Surface layers
- Corrosion or pitting
- Welds and heat-affected zones
- Etched microstructures

It is usually fast, relatively simple and covers a larger area than SEM. It is therefore very useful as a first characterization step because it helps decide where to perform more detailed characterization.

2.26 Limitations of LOM

The main limitations are,

- Limited resolution due to diffraction
- Strong dependence on sample preparation
- Limited depth of field
- Interpretation depends on contrast mechanism
- Cannot directly identify elemental composition
- Cannot directly identify crystal phases

LOM can show that different regions exist, but it usually cannot tell their exact chemistry or crystal structure. For that, SEM-EDS or XRD may be needed.

2.27 LOM compared with SEM

LOM uses visible light, while SEM uses electrons. Because electrons have a much shorter wavelength than visible light, SEM can achieve much higher resolution.

LOM is better for fast overview and larger areas. SEM is better for fine surface details, high magnification, fracture surfaces and small particles.

The main comparison is,

LOM $\Rightarrow$ overview and general microstructure

SEM $\Rightarrow$ higher resolution and fine surface detail

LOM is faster and simpler, but SEM gives more detail and can be combined with EDS for elemental analysis.

3 X-ray diffraction

X-ray diffraction, XRD, is a characterization method used to study crystalline materials. It is mainly used to identify crystalline phases, determine lattice parameters, estimate phase fractions, analyse residual stresses, and study crystallite size and microstrain from peak broadening.

The main idea is,

XRD measures how X-rays diffract from lattice planes in a crystalline material.

Since the wavelength of X-rays is of the same order as interatomic distances, X-rays can interfere constructively when scattered by the periodic arrangement of atoms in a crystal. This constructive interference produces diffraction peaks.

XRD is especially useful because it gives information about crystal structure and phases, not just elemental composition. This is a key difference from EDS, which mainly gives elemental information.

3.1 General properties of X-rays

X-rays are a form of electromagnetic radiation. They have no charge, no rest mass, and travel at the speed of light in vacuum.

Typical X-ray wavelengths are in the range,

$$0.01 \text{ nm} \lesssim \lambda \lesssim 10 \text{ nm}$$

For diffraction experiments, the most useful wavelengths are around the size of atomic plane spacings, usually around,

$$\lambda \sim 0.1 \text{ nm}$$

This is important because diffraction is only useful when the wavelength is comparable to the spacing between scattering centres. In crystals, these scattering centres are atoms arranged in periodic lattice planes.

The photon energy is related to wavelength by,

$$E = \frac{hc}{\lambda}$$

where E is the photon energy, h is Planck's constant, c is the speed of light and λ is the wavelength.

Thus,

$$\lambda \downarrow \Rightarrow E \uparrow$$

Shorter wavelength X-rays have higher energy and generally penetrate deeper into many materials.

3.2 X-ray generation

X-rays are commonly generated in an X-ray tube. The basic process is,

electron emission $\rightarrow$ electron acceleration $\rightarrow$ electron impact on anode $\rightarrow$ X-ray generation

Electrons are emitted from a cathode, often by thermionic emission from a heated filament. These electrons are accelerated by an applied voltage U and hit a metal anode. When the electrons are decelerated or interact with atoms in the anode, X-rays are produced.

Most of the electron energy is converted into heat, so the anode must be cooled. High intensity X-ray sources may use rotating anodes or liquid anodes to reduce overheating and anode damage.

There are two main types of X-ray radiation produced in an X-ray tube:

Bremsstrahlung radiation + characteristic radiation

3.3 Bremsstrahlung radiation

Bremsstrahlung means braking radiation. It is produced when high-energy electrons are decelerated in the electric field of atoms in the anode.

Because the electrons can lose different amounts of energy, Bremsstrahlung produces a continuous X-ray spectrum.

The maximum possible X-ray photon energy is equal to the kinetic energy of the incoming electron,

$$E_{\max} = eU$$

where e is the elementary charge and U is the acceleration voltage.

The corresponding minimum wavelength is,

$$\lambda_{\min} = \frac{hc}{eU}$$

This means that increasing the acceleration voltage gives access to shorter wavelength X-rays.

The important interpretation is,

$$U \uparrow \Rightarrow E_{\max} \uparrow \Rightarrow \lambda_{\min} \downarrow$$

Bremsstrahlung is usually not ideal for XRD because it contains many wavelengths. XRD normally needs radiation that is close to monochromatic, so characteristic X-rays and filtering are important.

3.4 Characteristic X-rays

Characteristic X-rays are produced when an incoming electron has enough energy to remove an inner-shell electron from an atom in the anode.

The process is,

inner-shell electron removed $\rightarrow$ core hole created $\rightarrow$ outer electron fills hole $\rightarrow$ X-ray photon emitted

The emitted X-ray energy equals the difference in binding energy between the two shells,

$$E_{X-ray} = E_{\text{higher shell}} - E_{\text{lower shell}}$$

Because the binding energies are specific to the element, the emitted X-rays are characteristic of the anode material.

Common characteristic lines are,

$$K_{\alpha 1}, \quad K_{\alpha 2}, \quad K_{\beta}$$

A K_{α} line occurs when an electron fills a vacancy in the K-shell from an L-shell. A K_{β} line occurs when an electron fills a vacancy in the K-shell from a higher shell, usually an M-shell.

For many anodes, $K_{\alpha 1}$ has approximately twice the intensity of $K_{\alpha 2}$. This is why diffraction peaks can sometimes appear slightly asymmetric or split if both components are present.

Common X-ray sources include,

$$\text{Cu } K_{\alpha}, \quad \text{Co } K_{\alpha}, \quad \text{Cr } K_{\alpha}, \quad \text{Mo } K_{\alpha}$$

The choice of source matters because different wavelengths give different peak positions and different penetration depths.

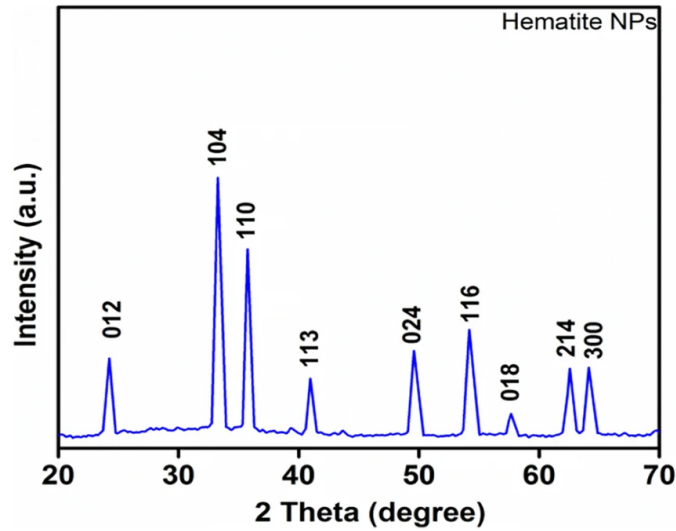


Figure 4: XRD pattern of synthesised iron oxide (α -Fe₂O₃) nanoparticles

3.5 X-ray attenuation

When X-rays pass through matter, their intensity decreases due to absorption and scattering. This reduction in intensity is called attenuation.

The intensity follows the Lambert-Beer law,

$$I(x) = I_0 e^{-\mu x}$$

where I_0 is the incoming intensity, $I(x)$ is the intensity after travelling a distance x , and μ is the linear attenuation coefficient.

The attenuation coefficient depends on the material and on the X-ray wavelength or energy. It is related to the mass attenuation coefficient κ by,

$$\mu = \rho \kappa$$

where ρ is the density.

The general trend is,

higher atomic number $\Rightarrow$ stronger attenuation

and,

higher X-ray energy $\Rightarrow$ lower attenuation, in general

There are also absorption edges. These occur when the X-ray photon energy becomes large enough to eject a core electron from a specific shell. At an absorption edge, attenuation increases suddenly.

3.6 Filtering of X-rays

Filtering is used to remove unwanted wavelengths from the X-ray spectrum. The goal is to obtain radiation that is as monochromatic as possible.

A common example is using a nickel filter for copper radiation. The filter strongly absorbs Cu K_β radiation while transmitting most of the Cu K_α radiation.

The idea is,

choose filter absorption edge between K_α and K_β

This allows the unwanted K_β component to be reduced.

For oral exam purposes, the key point is:

Filtering improves diffraction patterns by reducing unwanted wavelengths.

3.7 Diffraction and constructive interference

X-ray diffraction occurs because X-rays are elastically scattered by electrons in atoms. Each atom can act as a scattering centre. In a crystal, atoms are periodically arranged, so the scattered waves can interfere with each other.

Constructive interference occurs when the path difference between scattered waves is an integer number of wavelengths,

$$\Delta s = n\lambda$$

where n is an integer.

Destructive interference occurs when the waves are out of phase, causing cancellation.

A crystal has many regularly spaced planes of atoms. When X-rays scatter from these planes, constructive interference occurs only at specific angles. These angles produce diffraction peaks.

3.8 Bragg's law

The most important equation in XRD is Bragg's law,

$$2d \sin \theta = n\lambda$$

where d is the lattice plane spacing, θ is the Bragg angle, n is the diffraction order and λ is the X-ray wavelength.

The measured angle in a diffractogram is usually 2θ , not θ . This is because the detector measures the total scattering angle between the incoming and diffracted beam.

The Bragg angle is,

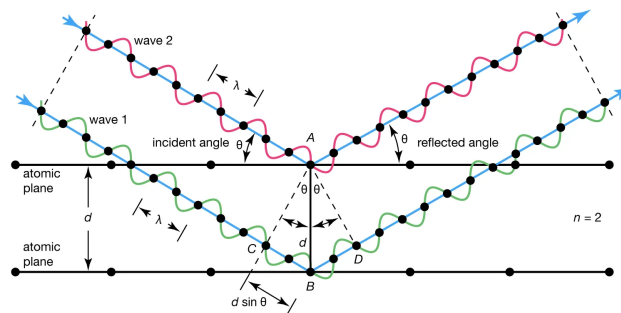
$$\theta = \frac{1}{2}(2\theta)$$

A diffraction peak appears when Bragg's law is satisfied for a specific family of lattice planes (hkl).

Important interpretation:

$$2\theta \uparrow \Rightarrow \theta \uparrow \Rightarrow d \downarrow$$

Thus, peaks at higher 2θ correspond to smaller lattice spacings.



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3.9 The diffracting plane is not necessarily the surface

In Bragg's law, the angle θ is defined relative to the diffracting lattice planes. These planes are not necessarily parallel to the sample surface.

The diffraction vector is perpendicular to the diffracting lattice planes. Therefore, XRD measures the spacing of lattice planes in the direction of the diffraction vector.

This is important in residual stress analysis, because tilting the sample changes which direction in the material is being measured.

A good oral answer is:

XRD does not simply measure the surface. It measures lattice spacings of planes that satisfy the diffraction condition, and the measured direction is related to the diffraction vector.

3.10 Lattice spacing and Miller indices

Each family of lattice planes is described by Miller indices (hkl) . For a cubic crystal, the lattice spacing is,

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

where a is the lattice parameter.

Combining this with Bragg's law gives,

$$2 \frac{a}{\sqrt{h^2 + k^2 + l^2}} \sin \theta = n\lambda$$

For first-order diffraction, $n = 1$, this can be rearranged to,

$$\sin^2 \theta = \frac{\lambda^2}{4a^2} (h^2 + k^2 + l^2)$$

Therefore, for cubic crystals,

$$\sin^2 \theta \propto h^2 + k^2 + l^2$$

This relation is useful for indexing unknown cubic diffraction patterns.

3.11 Single crystal and polycrystalline diffraction

A single crystal has only one orientation. Therefore, in a simple diffractometer geometry, only lattice planes with the correct orientation will satisfy Bragg's law. A single crystal may therefore show only one family of peaks, including higher orders.

A polycrystalline sample contains many grains with different orientations. If the grain orientations are randomly distributed and the grains are not too large, then for each possible set of lattice planes there will be some grains correctly oriented for diffraction.

Therefore, a powder or polycrystalline sample can show many possible reflections.

The important assumption in powder diffraction is,

$$\text{all grain orientations are statistically represented}$$

If this is not true, the sample has texture. Texture changes the relative peak intensities and can make phase analysis more difficult.

3.12 Powder diffraction

In powder diffraction, the diffractogram shows intensity as a function of scattering angle,

$$I = f(2\theta)$$

A peak occurs when a family of lattice planes satisfies Bragg's law.

The position of a peak gives information about lattice spacing. The intensity gives information about how much of that phase is present, but also depends on texture, structure factor, multiplicity and measurement conditions. The width and shape of a peak give information about crystallite size, microstrain and instrumental effects.

A useful summary is,

peak position $\rightarrow$ lattice spacing, phase, lattice parameter, macrostrain

peak intensity $\rightarrow$ phase fraction and crystallographic texture

peak width/shape $\rightarrow$ crystallite size and microstrain

3.13 Extinction rules for cubic lattices

Not every possible (hkl) reflection appears in every crystal structure. Some reflections are systematically absent due to destructive interference from the arrangement of atoms in the unit cell. These are called extinction rules.

For face-centred cubic, fcc, reflections are allowed when,

$$h, k, l \text{ are all even or all odd}$$

Examples of allowed fcc reflections are,

$$111, \quad 200, \quad 220, \quad 311, \quad 222$$

For body-centred cubic, bcc, reflections are allowed when,

$$h + k + l = \text{even}$$

Examples of allowed bcc reflections are,

$$110, \quad 200, \quad 211, \quad 220, \quad 310, \quad 222$$

These rules are useful because they allow fcc and bcc crystal structures to be distinguished from the sequence of diffraction peaks.

3.14 Indexing a cubic diffraction pattern

For an unknown cubic pattern, the procedure is:

1. Measure the peak positions 2θ .
2. Calculate $\theta = 2\theta/2$.
3. Calculate $\sin^2 \theta$ for each peak.
4. Divide all values by the smallest $\sin^2 \theta$.

5. Multiply by the smallest factor that gives allowed integers.
6. Compare the sequence with fcc or bcc extinction rules.
7. Assign the corresponding (hkl) values.

For cubic crystals, the sequence of $h^2 + k^2 + l^2$ values identifies the crystal structure.

For example, fcc often begins with,

$$3, 4, 8, 11, 12, \dots$$

while bcc often begins with,

$$2, 4, 6, 8, 10, 12, \dots$$

3.15 Phase identification

Qualitative phase analysis means identifying which crystalline phases are present in the material.

The process is,

measure diffraction pattern $\rightarrow$ find peak positions $\rightarrow$ compare with reference patterns

Reference patterns are stored in databases such as the JCPDS/ICDD Powder Diffraction File, PDF. These contain d -spacings and relative intensities for single-phase materials.

For reliable phase identification, all major measured peaks should be explained. Usually at least three peaks are needed to confidently confirm a phase.

A common oral exam trap is:

EDS identifies elements, while XRD identifies crystalline phases.

For example, EDS may show Fe and O, but XRD can help identify whether the phase is FeO, Fe₂O₃ or Fe₃O₄.

3.16 Quantitative phase analysis

Quantitative phase analysis estimates how much of each phase is present. It is based mainly on integrated peak intensities.

The integrated intensity is the area under a diffraction peak,

$$I_{int} = \int I(2\theta) d(2\theta)$$

A larger amount of a phase generally gives stronger peaks, but intensity also depends on other factors such as texture, structure factor, absorption, instrumental geometry and preferred orientation.

Therefore, quantitative phase analysis is more difficult than simply comparing peak heights.

For oral exam purposes:

Peak intensity can be used for quantitative phase analysis, but only carefully, because intensity is affected by texture and measurement conditions.

3.17 Composition from lattice parameter

If an alloying element changes the lattice parameter, XRD peak positions can be used to estimate composition.

First, the lattice spacing is calculated from Bragg's law,

$$d = \frac{n\lambda}{2 \sin \theta}$$

For a cubic material, the lattice parameter is then,

$$a = d_{hkl} \sqrt{h^2 + k^2 + l^2}$$

If the lattice parameter changes approximately linearly with composition, Vegard's law can be used,

$$a = a_0 + \beta c$$

where a_0 is the lattice parameter of the reference material, c is composition, and β describes how strongly the lattice parameter changes with composition.

The important interpretation is,

$$\text{peak shift} \rightarrow \text{change in } d_{hkl} \rightarrow \text{change in lattice parameter} \rightarrow \text{possible composition change}$$

However, peak shifts can also come from residual stress, so composition and stress effects must be separated carefully.

3.18 Residual stress analysis

Residual stresses change the spacing between lattice planes. XRD can measure this change as lattice strain.

The lattice strain for a given reflection (hkl) and measurement direction (ϕ, ψ) is,

$$\varepsilon_{\phi\psi}^{hkl} = \frac{d_{\phi\psi}^{hkl} - d_0^{hkl}}{d_0^{hkl}}$$

where $d_{\phi\psi}^{hkl}$ is the measured lattice spacing and d_0^{hkl} is the strain-free lattice spacing.

If the material is under tension, the lattice spacing increases in some directions and decreases in others due to elastic strain. If the material is under compression, the opposite shift occurs.

XRD residual stress analysis usually measures d_{hkl} at different sample tilts ψ . The change in lattice spacing with ψ is related to stress.

3.19 The $\sin^2 \psi$ method

The $\sin^2 \psi$ method is commonly used for residual stress analysis. In a biaxial stress state, the strain can be written as,

$$\varepsilon_{\phi\psi}^{hkl} = S_1^{hkl}(\sigma_1 + \sigma_2) + \frac{1}{2} S_2^{hkl} \sigma_\phi \sin^2 \psi$$

where S_1^{hkl} and S_2^{hkl} are X-ray elastic constants, σ_1 and σ_2 are principal stresses, σ_ϕ is the stress in the measured in-plane direction, and ψ is the sample tilt.

The key point is that strain is plotted as a function of $\sin^2 \psi$.

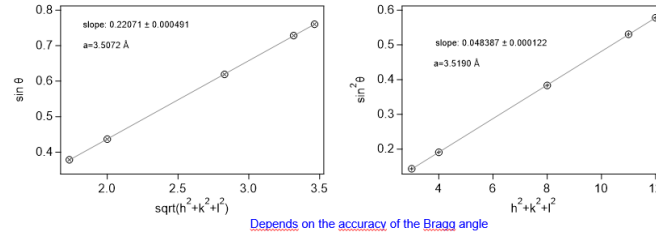
The slope is related to stress,

$$\text{slope} \propto \frac{1}{2} S_2^{hkl} \sigma_\phi$$

Therefore,

$$\text{slope of } d \text{ vs. } \sin^2 \psi \rightarrow \text{residual stress}$$

If the plot is linear, the assumptions of the method are more likely to be valid. Non-linear behaviour can indicate stress gradients, texture, phase gradients, shear stresses or other complexities.



3.20 Macrostrain and microstrain

Macrostrain is a uniform or direction-dependent change in lattice spacing over many grains. It mainly shifts the peak position.

$$\text{macrostrain} \rightarrow \text{peak shift}$$

Microstrain is a distribution of lattice spacings within grains. It can come from defects, dislocations, local stress variations or inhomogeneous strain.

$$\text{microstrain} \rightarrow \text{peak broadening}$$

The important distinction is:

$$\text{peak shift} \neq \text{peak broadening}$$

Peak shift is mainly connected to average lattice strain, while peak broadening is connected to variations in lattice spacing and finite crystallite size.

3.21 Information depth

XRD does not measure only the outermost surface. The measured signal comes from a certain information depth. The result is an absorption-weighted average over the diffracting volume.

For symmetrical X-ray geometry, the information depth can be written as,

$$\tau = \frac{\sin \theta \cos \psi}{2\mu}$$

where τ is the information depth, μ is the linear absorption coefficient, θ is the Bragg angle and ψ is the tilt angle.

This means that the information depth depends on material, wavelength, diffraction angle and tilt.

Important interpretation:

XRD gives absorption-weighted average values.

Therefore, for layered materials or surface treatments, the measured XRD signal may include contributions from different depths.

3.22 Grazing incidence diffraction

In grazing incidence XRD, the incident angle γ is kept small. This gives a shallower information depth and makes the method more surface sensitive.

For asymmetric grazing incidence geometry, the information depth is,

$$\tau = \frac{1}{\mu} \left(\frac{1}{\sin \gamma} + \frac{1}{\sin(2\theta - \gamma)} \right)^{-1}$$

where γ is the fixed incidence angle.

The main advantage is,

$$\gamma \downarrow \Rightarrow \tau \downarrow$$

Therefore, grazing incidence XRD is useful for thin films, coatings, expanded surface layers and near-surface residual stress or phase analysis.

3.23 Line profile analysis

Line profile analysis studies the shape and width of diffraction peaks.

Peak broadening can come from,

$$\begin{aligned} &\text{finite crystallite size} \\ &\text{microstrain} \\ &\text{instrumental broadening} \end{aligned}$$

Instrumental broadening must be corrected before crystallite size or microstrain are interpreted.

A common way to describe peak width is the full width at half maximum, FWHM. Another measure is the integral breadth,

$$\beta = \frac{I_{int}}{I_{max}}$$

where I_{int} is the integrated peak intensity and I_{max} is the maximum intensity.

Peak fitting is important because peak position, width and shape depend on how the peak is fitted. Common peak functions include Gaussian, Lorentzian and pseudo-Voigt functions.

3.24 Crystallite size broadening

Finite crystallite size causes peak broadening because very small coherently diffracting domains do not contain enough lattice planes to produce extremely sharp interference maxima.

The size here is not necessarily the particle size. It is the size of the coherently diffracting domain in the direction of the diffraction vector.

A useful relation is the Scherrer equation,

$$D = \frac{K\lambda}{\beta \cos \theta}$$

where D is the crystallite size, K is a shape factor, λ is the wavelength, β is the peak broadening in radians and θ is the Bragg angle.

The interpretation is,

$$\beta \uparrow \Rightarrow D \downarrow$$

Broader peaks can indicate smaller crystallites, but this is only valid after instrumental and strain broadening have been considered.

3.25 Microstrain broadening

Microstrain broadening is caused by variations in lattice spacing within grains. These variations produce a range of diffraction angles, so the peak becomes broader.

The relation between strain and peak shift/broadening can be approximated by,

$$\varepsilon = \frac{\Delta d}{d} = -\Delta\theta \cot \theta$$

The important idea is,

$$\text{variation in } d_{hkl} \rightarrow \text{variation in } \theta \rightarrow \text{broadened peak}$$

Microstrain is often associated with dislocations, defects, local stress variations and plastic deformation.

3.26 Limitations and common traps in XRD

Important limitations are,

- XRD mainly detects crystalline phases. Amorphous material gives broad diffuse humps rather than sharp peaks.
- Minor phases may be below the detection limit.
- Peaks from different phases can overlap.
- Texture can change peak intensities and complicate phase quantification.
- Peak shifts can come from both composition changes and residual stress.
- Peak broadening can come from size, microstrain and instrumental effects.
- XRD gives an average over the irradiated volume and information depth, not a highly local measurement.

A very important oral exam answer is:

XRD identifies crystalline phases from diffraction peak positions and intensities. EDS identifies elements. Therefore, XRD and EDS answer different questions and are often used together.

3.27 Good condensed oral answer

X-ray diffraction uses X-rays with wavelengths comparable to atomic spacings to produce constructive interference from crystal lattice planes. The condition for diffraction is given by Bragg's law, $2d \sin \theta = n\lambda$. The peak positions give lattice spacings and can be used for phase identification and lattice parameter determination. Peak intensities can be used for phase fractions and texture, while peak widths and shapes can be used to study crystallite size and microstrain. XRD can also measure residual stresses because elastic strain changes the lattice spacing. The method is powerful for crystalline materials, but it has limitations such as peak overlap, texture, detection limits, instrumental broadening and difficulty with amorphous phases.

4 Scanning Electron Microscopy and SEM-EDS

Scanning Electron Microscopy, SEM, is a microscopy method where a focused electron beam is scanned over the sample surface. The interaction between the electron beam and the sample produces signals such

as secondary electrons, backscattered electrons and characteristic X-rays. These signals are used to form images and to analyse composition.

The main idea is,

SEM uses electrons instead of light to obtain high-resolution surface information.

SEM is useful for observing surface morphology, topography, particles, coatings, fracture surfaces, pores, cracks, grains and different phases. When combined with Energy Dispersive X-ray Spectroscopy, EDS, SEM can also provide elemental composition.

4.1 Why electrons give higher resolution than light

Light optical microscopy is limited by the wavelength of visible light. Electrons can have much shorter wavelengths than visible light, so SEM can achieve much higher resolution.

The electron wavelength can be described by the de Broglie relation,

$$\lambda = \frac{h}{p}$$

where λ is the wavelength, h is Planck's constant and p is momentum.

For a non-relativistic electron,

$$p = mv$$

so,

$$\lambda = \frac{h}{mv}$$

Increasing the acceleration voltage increases the electron velocity and momentum, which decreases the electron wavelength.

The important interpretation is,

$$\text{electron wavelength} \ll \text{visible light wavelength}$$

Therefore, electrons can theoretically resolve much smaller features than optical microscopy. In practice, SEM resolution is also limited by probe size, lens aberrations, interaction volume, signal generation and sample effects.

4.2 SEM instrumentation

The main components of an SEM are,

- electron source or electron gun,
- condenser lenses,
- apertures,
- scan coils,
- objective lens,
- sample stage and chamber,
- detectors,
- vacuum system.

The electron gun generates electrons. The lenses focus the electrons into a small probe. The scan coils move the beam over the sample in a raster pattern. At each point, signals are collected by detectors and converted into image intensity.

The SEM image is therefore not formed all at once. It is built point by point as the beam scans across the surface.

4.3 Electron sources

Common SEM electron sources include,

tungsten filament, LaB₆, field emission gun, FEG

A tungsten filament is robust and relatively cheap, but has lower brightness and larger energy spread.

A LaB₆ source has higher brightness than tungsten and can give better resolution, but requires better vacuum.

A field emission gun, FEG, gives very high brightness and a smaller probe size. This is why FEG-SEMs can give high-resolution images, especially at low accelerating voltage.

The source affects,

brightness + probe size + resolution + beam current stability

4.4 Probe diameter and resolution

The electron probe diameter is one of the factors controlling SEM resolution. A smaller probe allows smaller features to be imaged.

A relation from the SEM slides is,

$$d_p \approx \sqrt{\frac{4i_p}{\beta\pi^2\alpha_f^2}}$$

where d_p is the probe diameter, i_p is the probe current, β is the brightness and α_f is the probe convergence angle.

The important trends are,

$$i_p \uparrow \Rightarrow d_p \uparrow$$

and,

$$\beta \uparrow \Rightarrow d_p \downarrow$$

A higher probe current gives a stronger signal, but it can increase spot size and reduce resolution. A brighter electron source can provide more current in a smaller probe.

For oral exam:

Good SEM imaging is a compromise between small probe size, sufficient signal, depth of field and sample stability.

4.5 Working distance

The working distance is the distance between the final lens/pole piece and the sample surface.

A shorter working distance usually improves resolution because the probe can be focused more tightly. However, a shorter working distance often gives a smaller depth of field.

A longer working distance gives more depth of field and more space for detectors, but can reduce resolution.

The general trend is,

short working distance → better resolution

long working distance → larger depth of field

The best working distance depends on whether the goal is high-resolution imaging, EDS analysis, topographic imaging or general overview.

4.6 Aperture size

Apertures limit the angular spread and current of the electron beam.

A smaller aperture reduces beam current and can improve depth of field and reduce aberrations, but the image may become noisy because fewer electrons reach the sample.

A larger aperture increases beam current and signal, but may increase aberrations and reduce depth of field.

The practical compromise is,

small aperture → better depth of field, lower signal

large aperture → higher signal, possibly lower resolution

4.7 Accelerating voltage

The accelerating voltage controls the energy of the primary electrons.

Low accelerating voltage gives,

smaller interaction volume + more surface sensitivity + less beam penetration

High accelerating voltage gives,

larger interaction volume + stronger signal + better X-ray excitation for EDS

For imaging surface details, low voltage can be useful because the signal comes from a shallower region. For EDS, the voltage must be high enough to excite the characteristic X-rays of the elements of interest.

A common rule is that the beam energy should be significantly higher than the X-ray line energy that needs to be excited. If the voltage is too low, some elements or lines will not be detected properly.

4.8 Interaction volume

When the electron beam enters the sample, the electrons scatter and generate signals from a finite volume. This is called the interaction volume.

The interaction volume depends mainly on,

- accelerating voltage,
- atomic number,
- density,
- sample composition,
- sample tilt.

The most important trends are,

$$\text{accelerating voltage} \uparrow \Rightarrow \text{interaction volume} \uparrow$$

and,

$$\text{atomic number} \uparrow \Rightarrow \text{interaction volume} \downarrow$$

Higher voltage electrons penetrate deeper. Materials with higher atomic number scatter electrons more strongly, so the interaction volume is usually smaller.

This is extremely important for EDS. The X-rays used for EDS are generated from a larger volume than the apparent beam spot. Therefore, the spatial resolution of EDS is worse than the imaging resolution of SEM.

For oral exam:

In SEM, the beam may be very small, but the signal can come from a larger interaction volume. This is especially important for BSE and EDS.

4.9 Signals generated in SEM

The main signals used in SEM are,

secondary electrons, SE
backscattered electrons, BSE
characteristic X-rays

Secondary electrons and backscattered electrons are used for imaging. Characteristic X-rays are used for EDS elemental analysis.

The signals originate from different depths. Secondary electrons are generated very close to the surface, while backscattered electrons and X-rays come from deeper in the interaction volume.

4.10 Secondary electrons

Secondary electrons, SE, are low-energy electrons produced by inelastic interactions between the primary beam and the sample.

They usually have low energy, on the order of a few eV, so only secondary electrons generated close to the surface can escape.

Therefore, SE imaging is very surface sensitive.

SE images mainly show,

topography + surface morphology

Edges, slopes, particles and rough regions often appear bright because more secondary electrons can escape or reach the detector.

SE is useful for,

- fracture surfaces,
- roughness,
- particles,
- pores and cracks,
- surface morphology,

- general high-resolution imaging.

For oral exam:

Secondary electrons give mainly topographic contrast because they come from very close to the surface.

4.11 Backscattered electrons

Backscattered electrons, BSE, are primary electrons that are elastically scattered back out of the sample.

They have higher energy than secondary electrons and can escape from deeper in the sample.

BSE images mainly show,

composition contrast + atomic number contrast

Heavier elements backscatter more electrons and therefore appear brighter. Lighter elements backscatter fewer electrons and appear darker.

The backscatter coefficient is,

$$\eta = \frac{n_{BSE}}{n_i}$$

where n_{BSE} is the number of backscattered electrons and n_i is the number of incident electrons.

The important trend is,

$$Z \uparrow \Rightarrow \eta \uparrow \Rightarrow \text{brighter BSE signal}$$

For oral exam:

BSE imaging is useful for seeing different phases because phases with higher average atomic number appear brighter.

4.12 SE compared with BSE

The main comparison is,

SE $\rightarrow$ topography and surface morphology

BSE $\rightarrow$ composition contrast and phase contrast

SE is more surface sensitive and usually gives better topographic detail. BSE comes from deeper and is more sensitive to atomic number.

A common example is a polished multi-phase sample. In SE mode, the image may mainly show scratches, relief or surface topography. In BSE mode, different phases may appear with different brightness due to different average atomic numbers.

This is one of the most likely oral exam questions.

4.13 Contrast formation in SEM

SEM contrast can come from several effects:

- topographic contrast,
- composition contrast,
- trajectory effect,
- electron number effect,

- edge effects,
- detector geometry.

Topographic contrast means that surface shape affects how many electrons escape and reach the detector.

Composition contrast means that different materials generate different signals, especially in BSE mode.

Detector position also matters. A feature facing the detector may appear brighter than a similar feature facing away from it.

Therefore, SEM brightness should not always be interpreted as composition. It may also be caused by topography or detector geometry.

4.14 Trajectory effect

The trajectory effect occurs because electrons leaving the sample follow different paths depending on where they are generated and the shape of the surface.

On rough surfaces, slopes and valleys can direct electrons toward or away from the detector. This creates shadowing-like contrast, similar to directional lighting in optical microscopy.

The important point is,

brightness can depend on geometry, not only material

For oral exam:

The trajectory effect means that SEM contrast depends on how emitted electrons travel to the detector. Surface slopes and detector position can strongly affect brightness.

4.15 Electron number effect

The electron number effect means that different surface positions allow different numbers of electrons to escape.

Raised features, edges or exposed surfaces often emit more detectable electrons. Depressions, pores or shielded regions may emit fewer electrons.

This contributes to topographic contrast in SE images.

For example,

edge $\rightarrow$ more electrons escape $\rightarrow$ brighter

and,

valley $\rightarrow$ fewer electrons escape $\rightarrow$ darker

4.16 Depth of field in SEM

SEM generally has a much larger depth of field than LOM. This is one reason SEM is useful for fracture surfaces and rough samples.

A relation from the slides is,

$$D_f = \frac{2R}{\tan \alpha_f}$$

where D_f is depth of field, R is resolution and α_f is the convergence semi-angle.

Smaller convergence angle gives larger depth of field,

$$\alpha_f \downarrow \Rightarrow D_f \uparrow$$

However, changing convergence angle also affects probe size, signal and resolution. SEM imaging is therefore always a compromise.

4.17 How to obtain a good SEM image

To obtain a good SEM image, the operator adjusts:

- focus,
- working distance,
- aperture size,
- accelerating voltage,
- probe current or spot size,
- astigmatism correction,
- brightness and contrast,
- detector choice,
- scan speed.

The correct settings depend on the goal.

For surface topography, SE imaging at lower voltage may be useful. For composition contrast, BSE imaging is useful. For EDS, the accelerating voltage and probe current must be high enough to generate enough X-rays.

4.18 Astigmatism

Astigmatism occurs when the electron beam is not focused equally in all directions. This makes the image appear stretched or blurred in one direction.

Astigmatism is corrected using stigmators. If astigmatism is not corrected, fine features will not be sharp even if the image appears roughly focused.

For oral exam:

Astigmatism makes the probe non-circular, so the image becomes directionally blurred. It must be corrected to obtain a sharp SEM image.

4.19 Charging

Charging occurs when electrons accumulate on a non-conductive or poorly grounded sample.

Charging can cause,

- brightness changes,
- image drift,
- image distortion,
- sudden movement of features,
- poor focus,
- unstable contrast.

Charging is common for ceramics, polymers, oxides, biological samples and poorly mounted samples.

Ways to reduce charging include,

- applying a conductive coating,
- reducing accelerating voltage,
- reducing beam current,
- improving grounding,
- using low-vacuum or environmental SEM modes.

For oral exam:

Charging happens when the incoming electron charge is not conducted away. It distorts the image and can make non-conductive samples difficult to image.

4.20 SEM sample preparation

SEM samples should generally be clean, dry, stable and compatible with vacuum.

Important preparation points are,

- the sample must fit in the chamber,
- loose particles should be removed,
- the surface should be representative,
- the sample should be electrically grounded,
- non-conductive samples may require coating,
- contamination should be minimized,
- fragile features should not be damaged during preparation.

Conductive coatings can be carbon, gold, platinum or other thin conductive layers. Carbon coating is often preferred when EDS is needed, because heavy metal coatings can introduce extra peaks or obscure low-energy X-rays.

However, carbon coating can affect carbon analysis.

This gives an important trade-off:

coating reduces charging, but can affect composition analysis

4.21 Types of SEM

Different SEMs have different electron sources and detector systems.

A field emission SEM usually gives higher resolution and better low-voltage performance than a simple tungsten filament SEM.

Common detectors include,

- Everhart-Thornley SE detector,
- in-lens SE detector,

- solid-state BSE detector,
- EDS detector.

The detector choice strongly affects contrast. Therefore, when interpreting SEM images, it is important to know whether the image was taken using SE or BSE.

4.22 SEM-EDS principle

Energy Dispersive X-ray Spectroscopy, EDS, is used in the SEM to analyse elemental composition.

When the electron beam hits the sample, it can eject an inner-shell electron from an atom. This creates a vacancy. An electron from a higher energy shell can then fill the vacancy, and the energy difference is emitted as a characteristic X-ray.

The process is,

electron beam $\rightarrow$ inner-shell ionization $\rightarrow$ electron transition $\rightarrow$ characteristic X-ray

Each element has characteristic X-ray energies, so measuring the X-ray energies allows the elements to be identified.

For oral exam:

EDS identifies elements by measuring the energies of characteristic X-rays generated by the electron beam.

4.23 Spectroscopy

Spectroscopy is the use of absorption, emission or scattering of electromagnetic radiation to study the chemical identity of matter or physical processes.

In SEM-EDS, the relevant radiation is emitted X-rays. The detector measures the energy and number of X-ray photons.

The EDS spectrum is usually shown as,

counts vs. X-ray energy

Each peak corresponds to an X-ray line from an element.

4.24 K, L and M lines

Characteristic X-ray lines are named according to the shell where the vacancy is filled.

A K-line means the final vacancy is in the K shell. An L-line means the final vacancy is in the L shell. An M-line means the final vacancy is in the M shell.

Examples are,

$$K_{\alpha} : L \rightarrow K$$

$$K_{\beta} : M \rightarrow K$$

For light and medium elements, K-lines are often used. For heavier elements, K-lines may be too high in energy, so L-lines or M-lines may be used instead.

A single element can therefore produce several peaks in an EDS spectrum.

4.25 EDS compared with WDS

EDS measures the energy of X-ray photons directly.

The photon energy is,

$$E = \frac{hc}{\lambda}$$

Wavelength Dispersive Spectroscopy, WDS, separates X-rays by wavelength using diffraction,

$$n\lambda = 2d \sin \theta$$

EDS is fast and convenient because it can collect many X-ray energies at once. WDS has better spectral resolution and can separate overlapping peaks better, but it is slower and more specialized.

The comparison is,

EDS → fast, common, lower spectral resolution

WDS → slower, higher spectral resolution

4.26 EDS microanalysis modes

EDS can be used in different modes:

- point analysis,
- area analysis,
- line scan,
- elemental mapping.

Point analysis measures the spectrum from a selected point, but the analysed volume is still determined by the interaction volume.

Area analysis averages over a selected region.

A line scan shows how element concentration changes along a line.

An elemental map shows the spatial distribution of selected elements.

For oral exam:

EDS maps show elemental distribution, but their spatial resolution is limited by the X-ray interaction volume.

4.27 Quantitative EDS

Quantitative EDS estimates the concentration of elements from X-ray peak intensities.

A basic comparison uses the ratio between the intensity of element A in the sample and in a reference,

$$k_A = \frac{I_A^{sample}}{I_A^{ref}}$$

The concentration is then corrected using ZAF corrections,

$$c_A = k_A \cdot Z \cdot A \cdot F \cdot c_A^{ref}$$

where Z is the atomic number correction, A is the absorption correction and F is the fluorescence correction.

The corrections are needed because the measured X-ray intensity does not depend only on concentration. It also depends on how electrons interact with the sample, how X-rays are absorbed before escaping, and whether secondary fluorescence occurs.

For oral exam:

EDS quantification is not just peak height. Corrections are needed because X-ray generation and absorption depend on the sample matrix.

4.28 EDS limitations

EDS has several important limitations:

- EDS gives elemental composition, not direct phase identification.
- Light elements are difficult to measure accurately.
- Peak overlaps can cause misidentification.
- The interaction volume limits spatial resolution.
- Surface contamination affects results.
- Rough surfaces can give inaccurate quantification.
- Thin coatings can be difficult because the substrate may also contribute signal.
- Carbon or oxygen peaks may come from contamination, resin, coating or oxidation.

The most important oral exam point is,

EDS identifies elements, not crystal structure.

Therefore, EDS cannot directly distinguish between different phases with the same elements. For example, it may detect Fe and O, but it cannot reliably distinguish between FeO, Fe₂O₃ and Fe₃O₄ without additional information such as XRD.

4.29 Carbon and oxygen in EDS

Carbon and oxygen peaks must be interpreted carefully.

Carbon can come from,

- carbon coating,
- contamination,
- mounting resin,
- hydrocarbons in the SEM chamber,
- the actual sample.

Oxygen can come from,

- oxides,
- surface oxidation,
- contamination,

- resin or polishing residues,
- the actual sample.

Therefore, detecting carbon or oxygen does not automatically mean they are part of the intended phase.

4.30 Thin coatings and small particles

Thin coatings and small particles are difficult in EDS because the X-ray signal comes from an interaction volume.

If the coating is thinner than the interaction volume, the EDS signal may include both coating and substrate.

Similarly, if a particle is smaller than the interaction volume, the measured spectrum may include the surrounding matrix.

The important point is,

$$\text{beam spot size} \neq \text{EDS spatial resolution}$$

Even if the beam is focused on a small feature, the X-rays can come from a larger volume.

To improve analysis of small features, one can use lower accelerating voltage, smaller beam current, careful sample preparation and appropriate standards, but the voltage must still be high enough to excite the required X-ray lines.

4.31 Combining SE, BSE and EDS

SE, BSE and EDS answer different questions.

SE → What does the surface look like?

BSE → Are there composition or phase contrast differences?

EDS → Which elements are present?

A good workflow is,

1. Use SE to observe morphology and surface features.
2. Use BSE to find phases or regions with different average atomic number.
3. Use EDS to analyse the elemental composition of selected regions.
4. Use XRD if crystalline phase identification is needed.

4.32 SEM compared with LOM

LOM uses visible light and lenses. SEM uses electrons and electromagnetic lenses.

LOM is useful for fast overview and larger areas. SEM gives higher resolution, greater depth of field and better imaging of fine surface features.

The comparison is,

LOM → overview and general microstructure

SEM → higher resolution and fine surface detail

However, SEM can be narrow-sighted because it often images a very small region. Therefore, SEM observations should be connected to larger-scale observations from LOM or other methods.

4.33 Good condensed oral answer for SEM

SEM uses a focused electron beam that scans over the sample surface. The electron beam interacts with the sample and produces signals such as secondary electrons and backscattered electrons. Secondary electrons are low-energy and come from near the surface, so they mainly give topographic contrast. Backscattered electrons are higher-energy primary electrons scattered out of the sample, and they mainly give composition or atomic number contrast. The SEM image depends on accelerating voltage, working distance, aperture size, probe current, astigmatism correction, detector type and sample preparation. The interaction volume is important because the signal does not always come only from the beam spot.

4.34 Good condensed oral answer for SEM-EDS

SEM-EDS uses the electron beam to generate characteristic X-rays from the sample. When an inner-shell electron is removed, an outer-shell electron fills the vacancy and emits an X-ray with an element-specific energy. The EDS detector measures these energies and produces a spectrum of counts versus energy. EDS can identify and estimate the concentration of elements, but it does not directly identify phases. It is limited by peak overlaps, poor accuracy for light elements, interaction volume effects, surface contamination and the need for matrix corrections in quantitative analysis. Therefore, EDS is often combined with BSE imaging and XRD.

5 Mechanical testing and Vickers hardness

Mechanical testing is used to measure how a material responds to applied mechanical loads. In materials characterization, mechanical testing is important because it connects the observed microstructure to the actual properties of the material.

The main idea is,

microstructure $\rightarrow$ mechanical response $\rightarrow$ material performance

Mechanical testing can include tensile testing, compression testing, bending, impact testing, fatigue testing, creep testing and hardness testing. In this course, the most relevant mechanical testing method is Vickers hardness testing.

5.1 Purpose of mechanical testing

The purpose of mechanical testing is to quantify properties such as,

- stiffness,
- strength,
- hardness,
- ductility,
- toughness,
- fatigue resistance,
- wear resistance.

Mechanical testing is not just about obtaining a number. The result should be related to the microstructure of the material. For example, grain size, phase distribution, porosity, heat treatment, precipitates, residual stresses and defects can all affect the measured mechanical properties.

For oral exam purposes, the important point is,

mechanical properties are controlled by microstructure

Therefore, mechanical testing should often be combined with microscopy, SEM-EDS or XRD.

5.2 Basic stress and strain

In mechanical testing, stress describes the force applied per area. Engineering stress is defined as,

$$\sigma = \frac{F}{A_0}$$

where F is the applied force and A_0 is the original cross-sectional area.

Engineering strain describes the relative change in length,

$$\varepsilon = \frac{\Delta L}{L_0}$$

where ΔL is the change in length and L_0 is the original length.

In the elastic region, stress and strain are related by Young's modulus,

$$E = \frac{\sigma}{\varepsilon}$$

where E is the elastic modulus.

The important interpretation is,

$$E \rightarrow \text{stiffness}$$

A high Young's modulus means that the material resists elastic deformation. It does not necessarily mean that the material is strong or tough.

5.3 Strength, ductility and toughness

The yield strength is the stress where plastic deformation begins. Below the yield strength, deformation is mainly elastic and reversible. Above the yield strength, permanent plastic deformation occurs.

The ultimate tensile strength is the maximum engineering stress reached during a tensile test.

Ductility describes how much plastic deformation the material can undergo before fracture. It is often measured as elongation at fracture or reduction in area.

Toughness describes the ability of a material to absorb energy before fracture. In a stress-strain curve, toughness is related to the area under the curve.

The difference between strength, ductility and toughness is important:

$$\text{strength} \neq \text{ductility} \neq \text{toughness}$$

A material can be strong but brittle. A material can also be ductile but not very strong.

For oral exam purposes:

Hardness is resistance to local plastic deformation, not the same as toughness.

5.4 Hardness testing

Hardness testing measures the resistance of a material to local plastic deformation. In indentation hardness testing, an indenter is pressed into the surface with a known force, and the size of the indentation is measured.

The general idea is,

smaller indentation $\rightarrow$ harder material

larger indentation $\rightarrow$ softer material

Hardness is useful because it is relatively fast, simple and local. It can be used to compare materials, check heat treatments, measure hardness profiles, study coatings and relate microstructure to mechanical response.

5.5 Vickers hardness principle

Vickers hardness uses a diamond pyramid indenter with a square base. The indenter is pressed into the material using a known load. After unloading, the two diagonals of the square indentation are measured.

The average diagonal is,

$$d = \frac{d_1 + d_2}{2}$$

where d_1 and d_2 are the two measured diagonals.

The Vickers hardness number is,

$$HV = \frac{1.854F}{d^2}$$

where F is the applied load and d is the average indentation diagonal.

The constant 1.854 comes from the geometry of the Vickers diamond pyramid.

The important interpretation is,

$$HV \propto \frac{F}{d^2}$$

For the same load,

$$d \uparrow \Rightarrow HV \downarrow$$

and,

$$d \downarrow \Rightarrow HV \uparrow$$

Thus, harder materials give smaller indents.

5.6 Physical meaning of Vickers hardness

Vickers hardness measures the resistance of the material to local plastic deformation under the indenter. During indentation, the material below the indenter is plastically deformed. The size of the indent depends on how easily the material yields and flows plastically.

For metals, higher hardness often corresponds to higher yield strength, but this relation is only approximate. Hardness is not a direct universal measurement of strength.

A common approximate relation for some metals is,

$$HV \sim 3\sigma_y$$

or,

$$H \approx 3\sigma_y$$

where H is hardness and σ_y is yield strength. This is only a rough estimate and should not be treated as exact.

5.7 Why the surface must be prepared

Vickers hardness requires a flat and polished surface because the indentation diagonals must be measured accurately.

The surface should be,

flat + smooth + free from preparation artefacts

Bad preparation can cause unreliable hardness values. For example,

- roughness makes the diagonals difficult to measure,
- scratches can hide the indentation edges,
- a tilted surface gives a distorted indentation,
- grinding deformation can artificially increase hardness,
- pores or cracks near the indent can reduce the measured hardness,
- indents too close to each other can interact through their plastic zones.

For oral exam purposes:

Bad sample preparation gives bad hardness data.

5.8 Indent spacing and edge distance

Indents should not be placed too close to each other or too close to an edge. This is because each indent creates a plastically deformed zone around it.

If indents are too close, the deformation field from one indent can affect the next one.

If an indent is too close to an edge, the material can deform more easily, giving an artificially low hardness value.

The important point is,

each indent must represent undisturbed material

5.9 Microhardness and macrohardness

Vickers hardness can be performed with different loads.

Low loads are often called microhardness testing. This is useful for small features, coatings, thin layers, weld zones or individual phases.

Higher loads are used for bulk hardness measurements.

The choice of load matters because the indent size determines the volume of material being tested.

For small features,

indent size must be smaller than the feature

For coatings,

indent depth should be small compared with coating thickness

Otherwise, the substrate can affect the measured hardness.

5.10 Hardness profiles

Hardness profiles are used when hardness changes with position. This is common in,

- carburized layers,
- nitrided layers,
- welded materials,
- heat-affected zones,
- surface-treated materials,
- coatings,
- additively manufactured materials.

A hardness profile is made by placing a series of indents at increasing distance from the surface or across a region of interest.

The result can show how processing changed the material locally.

For example,

nitriding → hard surface layer

welding → different hardness in weld metal, HAZ and base metal

5.11 Relation between hardness and microstructure

Hardness is strongly affected by microstructure.

Important microstructural effects include,

- smaller grain size usually increases hardness,
- martensite is harder than ferrite or austenite,
- precipitates and carbides can increase hardness,
- cold work increases hardness by increasing dislocation density,
- porosity can reduce measured hardness,
- cracks or weak interfaces can reduce local resistance to indentation.

The main idea is,

$$\text{obstacle to dislocation motion} \uparrow \Rightarrow \text{hardness} \uparrow$$

For metals, strengthening mechanisms often increase hardness because they make plastic deformation more difficult.

5.12 Residual stress and mechanical testing

Residual stress is stress that remains in a material without external loading. It can come from heat treatment, welding, machining, grinding, coating deposition, additive manufacturing, shot peening, carburizing or nitriding.

Residual stress can affect mechanical testing results.

Tensile residual stress is often harmful because it helps cracks open and grow. It can reduce fatigue life and increase the risk of fracture.

Compressive residual stress at the surface is often beneficial because it opposes crack opening. It can improve fatigue resistance and wear performance.

The important relation is,

$$\sigma_{\text{effective}} = \sigma_{\text{applied}} + \sigma_{\text{residual}}$$

If the residual stress is tensile, the effective tensile stress is higher. If the residual stress is compressive, the effective tensile stress at the surface can be reduced.

Residual stress can also influence indentation. For example, tensile or compressive stresses can slightly affect the shape and size of the indentation, especially in thin films or surface-treated layers.

For oral exam purposes:

XRD can measure residual stress, while mechanical testing shows how it affects performance.

5.13 Limitations of hardness testing

Important limitations of Vickers hardness are,

- it measures local indentation resistance, not full mechanical behaviour,

- it does not directly measure toughness,
- surface preparation strongly affects the result,
- diagonal measurement errors affect hardness strongly because $HV \propto 1/d^2$,
- thin coatings can be affected by the substrate,
- rough or porous materials can give scattered values,
- anisotropic or multiphase materials can give different values depending on indent position.

A very important oral exam point is,

hardness is not the same as strength or toughness

Hardness can often be correlated with strength, but it cannot replace tensile, fracture or fatigue testing.

5.14 Good condensed oral answer

Vickers hardness testing measures the resistance of a material to local plastic deformation. A diamond pyramid indenter is pressed into the surface with a known load, and the two indentation diagonals are measured. The hardness is calculated using $HV = 1.854F/d^2$, where F is the applied load and d is the average diagonal. A smaller indent means a harder material. The surface must be flat and polished so that the diagonals can be measured accurately. Hardness is affected by microstructure, such as grain size, phases, precipitates, cold work, porosity and residual stress. It is useful for comparing materials and making hardness profiles, but it does not directly measure toughness or full mechanical performance.

6 X-ray and neutron topography

X-ray and neutron topography are visualization methods where penetrating radiation is used to obtain information from the surface or interior of a material. Unlike ordinary optical microscopy or SEM, these methods can provide information from below the surface because X-rays and neutrons can penetrate into matter.

The main idea is,

X-rays or neutrons pass through or diffract from the material

and the recorded intensity gives information about internal structure, defects, strain, orientation or density variations.

6.1 Why use X-rays or neutrons for visualization?

Light optical microscopy and SEM mainly give information from the surface. X-rays and neutrons can penetrate deeper into materials, which makes them useful for internal characterization.

They can be used to study,

- internal defects,
- cracks,
- pores,
- inclusions,

- density variations,
- crystal defects,
- strain fields,
- orientation variations,
- coatings and surface layers,
- bulk microstructure.

The important advantage is,

penetrating radiation $\rightarrow$ non-destructive internal information

6.2 X-ray interaction with matter

X-rays interact mainly with electrons in matter. The main interactions are,

- absorption,
- elastic scattering,
- inelastic scattering,
- diffraction.

For imaging and topography, contrast can come from differences in absorption, scattering or diffraction.

In general, X-ray absorption increases with atomic number and density. Therefore, heavy or dense regions often absorb X-rays more strongly than light regions.

The intensity after passing through a material follows the Lambert-Beer law,

$$I(x) = I_0 e^{-\mu x}$$

where I_0 is the incident intensity, $I(x)$ is the transmitted intensity after travelling distance x , and μ is the linear attenuation coefficient.

The important interpretation is,

$$\mu \uparrow \Rightarrow I(x) \downarrow$$

A region with higher attenuation appears darker in transmission imaging.

6.3 X-ray radiography and tomography

X-ray radiography records a 2D projection of the transmitted X-ray intensity through a sample.

The image contrast depends on how much the X-rays are attenuated along each path through the material.

Radiography is useful for detecting,

- internal cracks,
- pores,

- voids,
- inclusions,
- density differences,
- lack of fusion defects in additive manufacturing,
- casting defects.

X-ray tomography, or CT, records many radiographs from different angles and reconstructs a 3D volume. The difference is,

radiography $\rightarrow$ 2D projection

tomography $\rightarrow$ 3D reconstruction

Tomography is especially useful when the position and shape of internal defects are important.

6.4 X-ray diffraction topography

X-ray diffraction topography uses diffracted X-rays to image variations in a crystal. Instead of only measuring a diffraction peak, the spatial distribution of diffracted intensity is recorded.

This can reveal defects and distortions in single crystals or large grains.

X-ray diffraction topography is sensitive to,

- dislocations,
- subgrain boundaries,
- strain fields,
- crystal curvature,
- orientation changes,
- defects that disturb the lattice.

The principle is related to Bragg's law,

$$2d \sin \theta = n\lambda$$

If the local lattice spacing d or orientation changes, the local diffraction condition changes. This changes the intensity recorded in the topographic image.

The important idea is,

local strain or orientation change $\rightarrow$ local change in diffracted intensity

6.5 Difference between XRD and X-ray topography

Conventional XRD usually gives an averaged diffraction pattern from the irradiated volume. It tells which phases are present, lattice spacings, residual stress, texture or crystallite size.

X-ray topography gives spatial information. It shows where in the sample the diffraction condition changes.

The comparison is,

XRD $\rightarrow$ average crystallographic information

X-ray topography $\rightarrow$ spatial map of crystal defects or strain

Thus, X-ray topography is more of an imaging method, while conventional XRD is more of a diffraction analysis method.

6.6 Neutron interaction with matter

Neutrons are different from X-rays because they have no electric charge and interact mainly with atomic nuclei and magnetic moments, not with electron clouds.

This gives neutron methods some special properties.

Neutrons can penetrate deeply into many materials, including metals. They can also be very sensitive to light elements such as hydrogen, which are difficult to detect with X-rays.

The important difference is,

X-rays interact mainly with electrons

neutrons interact mainly with nuclei and magnetic moments

Therefore, X-ray contrast often increases with atomic number, while neutron contrast does not simply follow atomic number.

6.7 Neutron imaging

Neutron imaging is similar in concept to X-ray imaging, but uses neutrons instead of X-rays. The transmitted neutron intensity depends on how the material attenuates or scatters neutrons.

Neutron imaging can be useful for,

- hydrogen-containing materials,
- water or moisture inside materials,
- polymers,
- batteries and fuel cells,
- internal defects in metals,
- thick components,
- magnetic materials.

Because neutrons can penetrate deeply, neutron imaging is useful for bulk and internal characterization.

6.8 Neutron diffraction and residual stress

Neutron diffraction can also be used to measure lattice spacing and residual stress, similar to XRD.

The same basic diffraction condition applies,

$$2d \sin \theta = n\lambda$$

The difference is that neutrons penetrate much deeper than laboratory X-rays. Therefore, neutron diffraction can measure residual stress inside bulk components, while XRD is more surface-sensitive.

The comparison is,

XRD residual stress $\rightarrow$ near-surface information

neutron diffraction residual stress $\rightarrow$ bulk/internal information

This makes neutron diffraction useful for thick parts, welds and engineering components.

6.9 Advantages of X-ray and neutron topography

Advantages include,

- non-destructive internal characterization,
- information from below the surface,
- sensitivity to internal defects,
- possibility of 3D imaging using tomography,
- strain and defect information using diffraction contrast,
- neutron sensitivity to light elements and magnetic structures.

6.10 Limitations of X-ray and neutron topography

Limitations include,

- equipment can be expensive and specialized,
- neutron sources are less accessible than laboratory X-ray sources,
- spatial resolution may be lower than SEM or LOM,
- interpretation can be complex,
- X-ray contrast may be weak between materials with similar density or atomic number,
- diffraction topography often requires crystalline samples and suitable orientation,
- measurements can average through thickness unless tomography or diffraction geometry is used carefully.

6.11 Good condensed oral answer

X-ray and neutron topography are visualization methods that use penetrating radiation to obtain information from inside materials. X-rays interact mainly with electrons, so contrast often depends on density and atomic number. Neutrons interact mainly with nuclei and magnetic moments, so they can give different contrast and can be sensitive to light elements such as hydrogen. X-ray radiography gives 2D projection images, while tomography can reconstruct 3D internal structures. Diffraction topography uses changes in diffracted intensity to reveal lattice defects, strain fields and orientation variations. Compared with LOM and SEM, these methods are less surface-limited and can be non-destructive, but they require more specialized equipment and interpretation.

7 Choosing characterization methods and relating results to properties

A major part of materials characterization is choosing the correct method or combination of methods for a given problem. No single method gives all information. Each method answers a different question.

The main idea is,

choose the method based on the question

For oral exams, it is not enough to describe each method separately. It is also important to explain why one method is chosen instead of another, and how different methods can be combined.

7.1 What each method mainly tells

A useful summary is,

LOM → overview microstructure

SEM-SE → surface topography and morphology

SEM-BSE → composition or phase contrast

EDS → elemental composition

XRD → crystalline phases and lattice parameters

Vickers hardness → resistance to local plastic deformation

X-ray/neutron imaging → internal defects or bulk information

Each method gives a different type of information, so they are often used together.

7.2 LOM

Light Optical Microscopy is usually used first because it gives a fast overview over a relatively large area.

LOM is useful for,

- grain structure,
- phase distribution,
- pores and cracks,
- coating thickness,
- weld zones,
- etched microstructures,
- corrosion and pitting.

Advantages:

- fast,
- relatively simple,
- large field of view,
- good for overview,
- useful after etching.

Limitations:

- limited resolution due to diffraction,
- depends strongly on sample preparation,
- cannot directly identify elements,
- cannot directly identify crystal phases.

Use LOM when the question is,

What does the general microstructure look like?

7.3 SEM

SEM is used when higher resolution or better depth of field is needed.

SEM-SE is useful for surface morphology and topography. SEM-BSE is useful for seeing different phases or regions with different average atomic number.

Advantages:

- high resolution,
- large depth of field,

- good for rough surfaces,
- good for particles, fracture surfaces and fine features,
- can be combined with EDS.

Limitations:

- small field of view can be narrow-sighted,
- sample must be vacuum compatible,
- non-conductive samples can charge,
- contrast can come from both topography and composition,
- sample preparation and detector choice strongly affect interpretation.

Use SEM when the question is,

What does the surface or fine microstructure look like?

7.4 EDS

EDS is used to identify which elements are present in a selected region.

Advantages:

- fast elemental analysis,
- can be combined directly with SEM imaging,
- can perform point analysis, area analysis, line scans and maps,
- useful for comparing phases or particles.

Limitations:

- identifies elements, not crystal phases,
- limited spatial resolution due to interaction volume,
- light elements are difficult,
- peak overlaps can cause errors,
- contamination can give misleading peaks,
- thin coatings and small particles can include signal from surrounding material.

Use EDS when the question is,

Which elements are present?

A key oral exam point is,

EDS identifies elements, not phases.

7.5 XRD

XRD is used to identify crystalline phases and obtain crystallographic information.

Advantages:

- identifies crystalline phases,
- gives lattice parameters,
- can estimate phase fractions,
- can measure residual stress,
- can estimate crystallite size and microstrain from peak broadening,
- can detect texture through intensity variations.

Limitations:

- mainly detects crystalline phases,
- amorphous material gives broad humps,
- minor phases may be below detection limit,
- peak overlap can complicate phase identification,
- texture affects intensities,
- gives averaged information over the irradiated volume.

Use XRD when the question is,

Which crystalline phases are present?

A key comparison is,

EDS → elements

XRD → crystalline phases

For example, EDS may detect Fe and O, but XRD can help distinguish between FeO, Fe₂O₃ and Fe₃O₄.

7.6 Vickers hardness

Vickers hardness is used to measure resistance to local plastic deformation.

Advantages:

- relatively simple,
- local measurement,
- useful for hardness profiles,
- useful for coatings and surface treatments if load is chosen correctly,

- can be related to microstructure and strengthening.

Limitations:

- does not directly measure toughness,
- does not give full stress-strain behaviour,
- depends strongly on surface preparation,
- thin coatings can be affected by the substrate,
- porous or heterogeneous materials can give scattered results.

Use Vickers hardness when the question is,

How resistant is the material to local plastic deformation?

7.7 X-ray and neutron imaging

X-ray and neutron imaging are useful when internal information is needed.

Advantages:

- non-destructive,
- can detect internal defects,
- tomography can give 3D information,
- neutrons can penetrate deeply and detect some light elements well.

Limitations:

- specialized equipment,
- lower spatial resolution than SEM in many cases,
- interpretation can be complex,
- contrast depends strongly on material and radiation type.

Use X-ray or neutron imaging when the question is,

Are there internal defects or bulk features?

7.8 Choosing methods for common cases

7.8.1 Unknown polished metal microstructure

A good method combination is,

LOM $\rightarrow$ SEM-BSE $\rightarrow$ EDS $\rightarrow$ XRD

LOM gives an overview of grains, phases and defects. SEM-BSE shows phase contrast. EDS checks elemental composition. XRD confirms crystalline phases.

7.8.2 Unknown particle or inclusion

A good method combination is,

SEM-SE + SEM-BSE + EDS

SE shows morphology. BSE shows whether it differs in atomic number from the matrix. EDS identifies the elements. If the phase must be identified, XRD may be needed, but small particles can be difficult to detect by bulk XRD.

7.8.3 Coating or surface treatment

A good method combination is,

cross-section LOM/SEM + EDS line scan/map + Vickers hardness profile + grazing incidence XRD

LOM or SEM measures thickness and morphology. EDS shows composition gradients. Hardness testing shows the mechanical effect of the treatment. Grazing incidence XRD can identify near-surface phases.

7.8.4 Fracture surface

A good method combination is,

SEM-SE + EDS if particles or corrosion products are present

SEM-SE is useful because fracture surfaces are rough and SEM has high depth of field. EDS can identify inclusions, oxides or contaminants on the fracture surface.

7.8.5 Residual stress problem

A good method combination is,

XRD residual stress + hardness + microstructure

XRD measures lattice strain and residual stress. Hardness testing shows changes in local mechanical response. LOM or SEM helps relate the stress state to microstructure, surface treatment or defects.

7.8.6 Porosity or internal defects

For surface-connected pores,

LOM + SEM

can be sufficient.

For internal pores,

X-ray tomography

is more suitable because it can reveal internal defects non-destructively.

7.9 How to analyse characterization data

For oral exam purposes, data analysis usually means understanding what the measured signal represents.

7.9.1 LOM data

In LOM, image contrast must be interpreted based on illumination mode and sample preparation.

Dark regions may be,

- grain boundaries,
- pores,
- cracks,
- etched phases,
- shadows,
- polishing artefacts.

Therefore, LOM images should not be interpreted without knowing the preparation and contrast mechanism.

7.9.2 SEM data

In SEM, brightness can come from topography or composition.

In SE images,

brightness $\sim$ surface topography and electron escape

In BSE images,

brightness $\sim$ average atomic number

Therefore, a bright region in SE is not necessarily a heavy element. It may simply be tilted toward the detector or located at an edge.

7.9.3 EDS data

In EDS spectra, peaks correspond to elements. Peak height or area is related to amount, but not directly equal to concentration without corrections.

Important checks are,

- Are peaks overlapping?
- Is the accelerating voltage high enough?
- Could carbon or oxygen come from contamination?
- Is the analysed feature larger than the interaction volume?
- Could the substrate contribute signal?

7.9.4 XRD data

In XRD,

peak position $\rightarrow d_{hkl}$, phase, lattice parameter, residual stress

peak intensity $\rightarrow$ phase fraction and texture

peak width $\rightarrow$ crystallite size, microstrain and instrumental broadening

Therefore, XRD interpretation should consider position, intensity and width, not only whether a peak is present.

7.9.5 Hardness data

In Vickers hardness,

$$HV = \frac{1.854F}{d^2}$$

A smaller indent gives higher hardness. However, scattered hardness values may indicate heterogeneous microstructure, porosity, poor surface preparation, cracking, or indents placed too close to edges or phase boundaries.

7.10 Microstructure–property relations

Characterization results become useful when they are related to material properties.

The main idea is,

processing $\rightarrow$ microstructure $\rightarrow$ properties $\rightarrow$ performance

7.10.1 Grain size

Smaller grains usually increase strength and hardness because grain boundaries obstruct dislocation motion.

grain size $\downarrow \Rightarrow$ strength and hardness $\uparrow$

This is related to Hall-Petch strengthening.

7.10.2 Phases

Different phases have different properties. For example, martensite is hard and strong but can be brittle. Austenite is usually more ductile. Ferrite is softer than martensite.

Therefore, phase identification by XRD and phase imaging by LOM/SEM are important for explaining hardness and strength.

7.10.3 Precipitates and carbides

Precipitates and carbides can strengthen a material by blocking dislocation motion.

precipitates/carbides $\rightarrow$ hardness increase

However, large brittle particles can also reduce toughness or act as crack initiation sites.

7.10.4 Porosity

Porosity usually reduces mechanical performance because pores reduce the load-bearing area and act as stress concentrators.

porosity $\uparrow \Rightarrow$ strength, fatigue resistance and toughness $\downarrow$

Porosity can be observed by LOM, SEM or X-ray tomography.

7.10.5 Cracks

Cracks are critical defects because they concentrate stress.

$$K = Y\sigma\sqrt{\pi a}$$

where K is the stress intensity factor, Y is a geometry factor, σ is applied stress and a is crack length.

A larger crack or higher stress increases the risk of fracture.

7.10.6 Residual stress

Residual stresses influence mechanical performance.

Tensile residual stress is usually harmful because it promotes crack opening.

Compressive residual stress at the surface is often beneficial because it reduces the effective tensile stress and can improve fatigue resistance.

$$\sigma_{\text{effective}} = \sigma_{\text{applied}} + \sigma_{\text{residual}}$$

XRD can measure residual stress by detecting changes in lattice spacing.

7.10.7 Crystallite size and microstrain

XRD peak broadening can indicate small crystallite size or microstrain.

Small crystallites and high defect density can increase hardness because they obstruct dislocation motion.

peak broadening $\rightarrow$ small crystallites and/or microstrain

This can be connected to strengthening, cold work, deformation or processing history.

7.10.8 Coatings and surface layers

Coatings and surface treatments are often used to improve wear resistance, corrosion resistance or fatigue resistance.

Important characterization questions are,

- How thick is the coating?
- Is it continuous?
- What elements are present?
- Which phases are present?
- Is the surface harder than the substrate?
- Are there cracks, pores or delamination?
- Are residual stresses present?

A useful combination is,

LOM/SEM cross-section + EDS + XRD + hardness profile

7.11 Good condensed oral answer

When choosing characterization methods, the method should match the question. LOM is good for overview microstructure, SEM-SE for surface topography, SEM-BSE for composition contrast, EDS for elemental composition, XRD for crystalline phases and residual stress, and Vickers hardness for resistance to local plastic deformation. These methods should often be combined because each has limitations. For example, EDS can show that Fe and O are present, but XRD is needed to identify the oxide phase. Similarly, hardness values become meaningful when related to microstructure, such as grain size, phases, porosity, precipitates and residual stress. The final goal is to connect characterization results to microstructure–property relations.